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Visualising and Predicting the Impact of Natural Disasters on Insurance Claims in the UK

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Abstract

This project develops a prototype decision support system to analyse and predict the impacts of natural disasters on insurance claims in the United Kingdom. The system was developed using R shiny to integrate Natural Disaster dataset, UK climate dataset, County Climate dataset and Insurance Claims dataset into an interactive dashboard. The key tab (modules) includes the Dashboard Overview, Data Explorer, Spatial Statistics, Predictive Insights and About & Download sections.

The Spatial analysis is integrated by utilising the Moran's I and the Moran scatterplots to detect the clustering in claims and residuals. The predictive modelling compared and evaluated the Generalised Linear Models and XGBoost (Poisson for counts and Tweedie for amounts) which highlights the trade-off between statistical interpretability and predictive flexibility.

The results shows that the GLM has captured spikes more effectively, whereas the XGBoost has produced a stable forecast in the zero heavy data but still struggles with extremes. The model is further implemented with stress test controls for insurers and policy makers to cast scenarios for forecasting. This study concludes that this prototype demonstrates feasibility, with possible future improvements that includes real claims, real time disaster and climate updates via API's, automated pipelines and cloud deployment for scalability.

Keywords: Natural Disasters, Insurance claims, Predictive Modelling, GLM, XGBoost, Spatial Analysis, R Shiny, Risk Management, Usability

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1. Introduction

1.1 Background and Motivation

Natural disasters have a profound and growing impact on the insurance sector, resulting in an increased surge of claims, heightened premiums rates and financial pressure on both insurers and policyholders. The (Swiss Re, 2024) reported that in 2023, natural disasters resulted in a global economic loss amounting to $280 billion, of which more than 50% were not covered by insurance. Recent data statistics from (The ABI, 2024) reveals that the total claims payout up to the third quarter of 2024 was £4.1 billion setting a record-breaking amount to be paid out in the first nine months of any year. Notably, in the third quarter of 2024, claims regarding damage to residence resulting from storms, significant rainfall, and frozen pipes accounted to £136 million, whereby marking the sixth consecutive quarter where weather related claims exceeded £100 million. Furthermore, this is a 6% increase compared to the £128 million disbursed in the same quarter of 2023.

The risk is expected to escalate. According to (Environment Agency, 2018) due to the impact of climate change, it is projected that there will be an increase in the total number of properties located in the region that will be vulnerable to flooding from rivers, sea or surface water, potentially reaching 8 million properties (increased from 6.3 million properties) by mid-century. Meaning one in four properties will be at risk of such flooding. These trends highlight the critical need for the insurance sector to improve its readiness, pricing frameworks and claims handling management strategies in response to the growing and escalating climate related risks.

Despite the extensive datasets on disaster, property risks and claim histories collected by insurers and government, there is frequent lack of integrated and user-friendly tools that can convert this raw data into practical insights. Studies by (Manfré et al., 2012a) highlights the idea of having geospatial database infrastructure to generate information which will be useful for preparedness and risk management strategies. Furthermore, as noted by (Kasaraneni, 2020a), machine learning techniques presents a transformative opportunity within the insurance industry for its ability to leverage powerful algorithms to analyse vast amounts of data which can help the insurers to gain deeper insight and deliver a more efficient customer centric insurance experience.

Given this potential, current industry tools often handle geospatial risk mapping and predictive claims modelling separately, failing to integrate them into a cohesive, interactive platform designed for real time analysis. This project aims to close the gap by developing an interactive dashboard that analyses and forecasts the effect of natural disasters and catastrophes on insurance claims within the United Kingdom. By leveraging historical data in disasters, information regarding insurance claims and geospatial mapping techniques, the application attempts to deliver practical insights concerning claim trends, regional risk assessments and financial consequences. The development of this dashboard represents a significant contribution to the ongoing efforts aimed at reducing the financial ricks connected to disasters faced in the UK, thereby facilitating to engage in proactive risk management, optimise resource distribution, and foster policy innovation in light of a changing climate.

This project is interdisciplinary by incorporating methodologies from data science, geospatial analytics and financial modelling. It is in alignment with the MSc Data Science (Fintech) program by utilising advanced technical and analytical skills and apply on a significant real-world issue. Modules such a Machine Learning and Applied Machine Learning will facilitate the formation of machine learning models whereas Data Visualisation and Big Data will assist in

developing a user-friendly interface. By merging the theoretical research with practical application, this project will satisfy not only the MSc program's requirements but also offers to valuable insights into industry practices, therefore, adhering to the QFC guidelines for academic integrity, rigor and relevance.

1.2 Problem Statement

The primary problem addressed in this project is an absence of integrated tools that merge geospatial analysis and predictive modelling for predicting insurance claims. The current industry practice frequently isolates the visualisation of disaster risk at regional levels from models used for predicting claims and hence restricts the ability of the insurer to evaluate their financial exposure. Such a fragmentation hinders the insurers from developing mitigation strategies for disaster preparedness, determining premiums and claim handling.

1.3 Aims and Objectives

1.3.1 Aims

The aim of this project is to develop an interactive dashboard enhanced by machine learning, designed to assist insurers and policy makers in the United Kingdom to visualise, analyse and forecast the financial repercussions of natural disasters on insurance claims. By integrating historical data on disasters and claims, geospatial analysis and predictive modelling, this project has the potential to yield a practical platform for enhanced preparedness, optimise resource distribution and improved risk mitigation strategies.

1.3.2 Objectives

To fulfil the aim, the project will set out to achieve the following specific objectives:

- Analyse historical data related to natural disaster and insurance claims.
    - Perform a comprehensive literature review to investigate the relationship between natural disasters and insurance claims, while identifying important patterns, challenges and emerging trends.
    - Gather and clean datasets pertaining to floods, storms, and related events, along with relevant insurance claims.
- Analyse the patterns and relationships present with the dataset.
    - Conduct an Exploratory Data Analysis (EDA) to reveal temporal, seasonal and geographic trends in claims patterns, including surges following disasters.
    - Assess the extent to which certain regions are adversely impacted by particular disasters and analyse regional risk assessments by using geospatial analytical tools.
    - Identify and extract relevant features in order to guide the predictive modelling phase.
- Develop and evaluate machine learning models aimed at estimating the volume and costs of claims.
    - Build, train and test machine learning models to forecast and predict future claim across a range of disaster scenarios.
    - Evaluate the model's accuracy by employing evaluation metrics and record the rationale for the choice of model.

- Design and implement an interactive dashboard for the analysis of disaster insurance
    - Develop an intuitive interface that includes filters for the time frame, geographical area and type of disaster.
    - Allow users to investigate patterns and produce predictors tailored to specific scenarios by incorporating interactive visualisations with the machine learning models that have been developed.
    - It is essential that the system architecture facilitates usability, scalability and the effective communication of insights.
- Test and evaluate the completed system.
    - Perform a comprehensive evaluation of the dashboard and machine learning models to determine their accuracy, usability and performance.
    - Enhance the system to tackle any limitations or areas that require improvement after taking into consideration of the feedback provided by the supervisor.
- Compile the findings and produce the final project report.
    - Compile a thorough find report that outlines the project's background, literature review, describe the system design and implementation, details of the testing methods, present an evaluation of the outcomes, draw conclusion and provide future recommendations.
    - Ensure that the report complies with academic standards, which include appropriate referencing, a clear presentation and a thoughtful analysis of project outcomes.

1.4 Research Questions and Hypotheses

1.4.1 Research Questions

The project is guided by the following research questions:

- In what ways can geospatial analysis and machine learning be integrated within an interactive platform to predict financial repercussions of natural disasters on insurance claims in the UK?
- To what degree can claims frequency and severity be precise and accurately predict by predictive models?
- How might an interactive dashboard allow insurers to better understand regional risk, better prepare against climate driven disasters and process claims effectively?

1.4.2 Working Hypotheses

Despite the project being mostly experimental, the project will function based on these working hypotheses.

- Hypothesis 1: Areas that have experienced a significant history of severe or frequent natural disasters will show a disproportionately high volumes and cost of insurance claims which indicates geographic vulnerability.
- Hypothesis 2: Predictive accuracy that are assessed through metrics like RMSE or MAE may be attained by machine learning models that are trained using historical data whereby making them an effective tool for practical risk forecasting with the insurance industry.
- Hypothesis 3: An interactive dashboard which combines geospatial visualisation with predictive analytics may enhance the user's abilities to investigate disaster related insurance risk scenarios when compared to static reports or non-interactive tools.

1.5 Methodology Overview

This project adheres to a systematic methodology comprising of four phases.

Data Collection and Preprocessing: Gather and refine datasets from EM-DAT, UK Met Office, Environment and Weather Agency. Standardise and aggregate geographical data to county level, time stamps and create features like lags and event indicators. Generate a simulated insurance claims dataset to fill in gaps and facilitate in controlled experiments.

System Design: Develop a modular based architecture that integrates machine learning with an interactive dashboard. Define data flows, model and caching mechanisms. Documenting the process using ERD, Use Case, Activity Diagram along with UI wireframes.

Implementation: Train the predictive models on claim counts and claim amounts considering time and geography splits to minimise leakage. Tune and save model, incorporate predictors onto the dashboard and incorporate the geospatial maps and EDA views.

Testing and Evaluation: Use evaluation metrics and error analysis for the modelling by county and natural disaster.

2. Literature Review

2.1 Impacts of Natural Disasters on the UK Insurance Industry

Natural disasters especially flooding and windstorms, have significantly impacted the United Kingdom's insurance sector in the recent years. Insurers are experiencing record volumes of claims and payout expenses due to unprecedented weather events, with various recent years attaining record breaking levels. As an instance, weather related damage claims reached an all-time high in 2023 of £573 million which is an increase of 36% compared to 2022 (The ABI, 2024), moreover the data revealed that while the number of claims has not dramatically risen, the severity (cost per claim) is rising which indicates that the average claim in 2023 amounted to approximately £6,235, which is an 11% increase from 2022. These patterns are primarily traced back to the series of severe storms and floods, partly attributed to climate change, to render these extreme weather "not feel so rare" in the UK (The Gurdian, 2024). It has been noticed by insurers that clusters of storms and heavy downpours have become increasingly common, hence putting pressure on their finances and operating capacities. Concurrently, insurance premiums charges have not been kept up with these escalating losses and when adjusted for inflation, average claim costs have risen while premium prices have fallen in real terms therefore suggesting that the competitive nature of the industry has thus far mitigated price hike (The ABI, 2024).

Various forms of natural disaster affect insurers in different manners. Historically, the most damaging destruction risk in the UK has been flooding, leading to fewer but costly claims meanwhile windstorms result in a greater number of low-cost claims. For instance, the floods during the winter of 2015-2016 (Storms Desmond , Eva and Frank) resulted in approximately 15,000 flood insurance policies were claimed, with estimated insured losses totally £1.3 billion (Artemis, 2016), The average compensation for each flooded home averaged about £50,000 while flooded businesses claimed about £135,000 (Artemis, 2016), highlighting the extreme destruction that floodwater can cause. Windstorms, by comparison, generally tend to cause less severe damage (such as roofs and fences) over a larger area, leading to numerous claims of low

costs claims. For example, the Storms Ciara and Dennis in February 2020 produced over 82,000 claims and an estimated payout of £363 million. With that amount, £214 million payout was responsible for flooding (the result was 3,350 domestic property flood claims, totalling an estimated £107 million, which means that the average cost of claim for each property was around £32,000), meanwhile a £149 million payout was responsible for windstorms (the result was £61,000 domestic property with windstorms damage claims, totalling an estimated £77 million which means that the average cost of claim for each property was around £1,300) (Walker, 2020).

This proves that repairing a flooded home significantly outweighed the costs of typical wind damage claim, likewise, a series of storms in February 2022, namely Dudley, Eunice, and Franklin led to 177,000 claims, with a total payout reaching £497 million, a scale that is comparable to the winter storms of 2020 (Storms Ciara, Dennis and Jorge) which incurred costs of approximately £543 million (Willard, 2022). These statistics highlight how clusters of storms can generate expensive insurance liabilities, even when individual wind claims are relatively minor. Beyond floods and winds, other types of natural disasters caused a surge in claims. For instance, an intense freeze, dubbed the “Beast from the East” in early 2018, caused record burst pipe claims of £194 million in just a single quarter. While the summer heatwave of 2018 which led the soil to dry out that led to quadrupling of subsidence claims with 10,000 claims totalling £64 million making the largest quarterly increase on record (Read, 2018). These occurrences illustrates that the Uk insurers are confronted with a range of climate related risks which ranges from floods and storms to extreme cold and heat of all which have potential to elevate both the number and cost of claims.

2.2 Predictive Models for Insurance Claims and Disaster Impacts

Predicting non-life insurance, a common assumption is to split it into claim frequency (how many claims) and severity (how large are the claims) explains (Laudagé et al., 2019). The data often get skewed with extreme weather conditions, which results in having many small claims with a few large ones involving significant complexity and severity explains (Araiza Carlos, 2023). Any methodology needs to honour these features to demonstrate an effective performance on unseen data and still remain comprehensible for control purposes.

(Wilson et al., 2024) claims that Generalised Linear Modelling (GLM) methods have been the standard industry practice for non-life insurance pricing for a long time. A study by (Lally & Hartman, 2015) showed that Poisson GLMs and popular alternatives like zero inflated Poisson and negative binominal GLMs systematically overestimated. This overestimation can have a serious problem and increase the risk of making poor pricing decisions. Furthermore, the study reported that Tweedie GLMs produced much better predictions compared to the Poisson GLMs. However, it was stated that even though there were significant performance gains, that Tweedie is more complex than Poisson, but this additional complexity should not be a barrier to implement Tweedie GLMs.

A study by (Pesantez-Narvaez et al., 2019) concluded that tree ensemble and gradient boosting model produced better results compared to linear regression models like logistic regression. When compared the XGBoost predictive performance was much higher than the logistic regression in the training sample but considerably poorer in the testing sample. The report showcased that XGBoost achieved much better discrimination than logistic regression for claim tendency which highlights gains when high dimensional behaviour features are available.

The monthly county panels often exhibit a heavy zero value with numerous months recording no claims and subsequent spikes following storms. A study by (So & Valdez, 2024) explains that an extension to the Tweedie family can tackle this issue by explicitly modelling the zero state and linking it to the mean by employing boosting techniques to capture non-linear effects. Furthermore, the imperial resulted that zero inflated Tweedie boosted trees outperformed traditional Tweedie GLMs. Simultaneously, another study by (Garrido et al., 2016) showed that the dependency between frequency and severity can skew the two-part fits and a joint specification within a GLM framework model frequently and conditional severity together which demonstrated effectiveness on insurance data. These aspects are directly relevant to disaster linked claims, which tends to be sparse during calm periods and cluster following the events of disasters.

2.3 Geospatial Analysis Framework for Disaster Risk and Insurance Claims

In the UK, flood risk maps or spatially aggregated risk data (the product of flood probability, exposure and vulnerability) are mostly used to inform flood defence investment policy and long-term risk planning explains (Bates et al., 2023). It is further elaborated that the workflow specifically generates the return period hazard layers, characterises the spatial dependence of floods, produces event footprints and integrates these with building exposure and dept damage (vulnerability) functions assess financial losses and loss exceedance curves. Both (Bates et al., 2023) and (Penning-Rowsell, 2021) stated that UK analysis are drawn from the national assessments like the NaFRA, CCRA and SoN, however, modelled values of annual average flood damage are between 2.06 and over 9.0 times the comparable flood losses as measured in terms of the compensation paid to flood victims by insurance companies (Penning-Rowsell, 2021). These studies make it clear that geospatial workflows test hazard layers, exposure inventories and vulnerability functions which are combined in a consistent pipeline.

Analysts apply spatial diagnostics to understanding the patterns in the hazard, exposure or claims. The common statistic found in the literature is the Global Moran's I for overall autocorrelation, Local Moran's I (LISA) for local clusters and spatial outliers and Getis-Ord Gi* for hot and cold spots. (Hassan et al., 2020) conducted a mapping of statistically significant flood clusters following the Cyclone Amphan by using both Local Moron's I and Gi* whereas (Röthlisberger et al., 2017) pinpoints flood exposure hotspots using Gi* as well as examining the sensitivity related to the distance bands and the aggregation units. Furthermore, recent research on Moran based autocorrelation functions by (Chen, 2021) has introduce methodological updates which emphases the appropriate contexts for using the global or local indicators.

2.4 Interactive Dashboard and Decision Support Tools in Insurance and Disaster Risk Analytics

(Mohanty & Karmakar, 2021)'s WebFRIS is a web-based tool that transforms flood information into maps at the village level, illustrating hazards, socio-economic vulnerabilities and overall risk through a straightforward category, with an interface that is tailored for non-technical users. A report by (Bakhtiari et al., 2023) reviews modern visualisation techniques for flooding and concluded that virtual reality (VR), augmented reality (AR) and digital twins are the primary tools used. It is further argued that most studies concentrate on preparedness and mitigation instead of a stronger connection to real time data and broader application throughout the whole risk management. (Nygren et al., 2024) developed an automated Python workflow that processes ICEYE SAR flood depth raster alongside global open data (DLR population, OpenStreetMap) to produce rapid, web-based impact maps which emphasises the necessity of evaluating uncertainty.

(Dong et al., 2020) examines the John Hopkins University COVID-19 dashboard which includes semi-automated feeds, frequent refresh and an open repository. It includes a model for real time high trust dashboard in rapidly changing situations.

Firstly, from (Mohanty & Karmakar, 2021)'s WebFRIS make use of the map first web interface which allows people with any technical training to use the system and ensure that outputs are presented in a clear manner. Subsequently, (Nygren et al., 2024) demonstrates that a repeatable pipeline standardises the input and generates consistency. Furthermore, (Bakhtiari et al., 2023) showcased that advanced visualisation techniques can provide value while also prioritising the visuals to actionable indicators and real time feeds. Lastly, (Dong et al., 2020) reported that the dashboard should be regarded as an operational product where it is important to document data sources and update times, maintain frequent refreshes and have clear transparency so that the users can trust what they see.

2.5 Gaps in Literature

In the context of the studies focused on the UK and the industry, the majority of this analysis indicated the impacts and risks at either the national or broad regional level. There is a scarcity of research that effectively integrates hazard layers, climate signals, exposure inventories and recorded claims at the county level on a consistent period of time, even though practical challenges in operation, including geocoding accuracy, varying boundaries and irregular frequencies for updates are noted, these are not frequently addressed in a manner that would facilitate routine workflow for insurers. Consequently, there exist a significant gap for a detailed and integrated dataset that can be synchronised space and time dimensions to enable measurement and comparison of local risk patterns.

In modelling, GLMs continue to serve as a benchmark, however they may overlook significant non-linear effects. In contrast, boosted trees enhance accuracy but often introduce challenges related to explainability and governance questions. Moreover, a limited number of studies have integrated spatial diagnostics during the modelling phase such as integrating neighbourhood structures or spatial characteristics with conventional predictors. Furthermore, data that have high frequency of zeros and are driven by events are common at monthly county level. Current methodology like zero-inflated Tweedie model is not extensively tested on the claims. Even the measure of results is restricted to calibrations, regional stability and error analysis by disasters or season are less commonly documented.

Currently, the tools for mapping and modelling remain isolated from one another. Numerous dashboards offer either exploratory or static maps which lacks scenario ready forecasts and displays model outputs without providing any clear connections to the underlying data slice, version of the model and appropriate uncertainties, there is an obvious requirement for an integrated county level data and models that illustrates uncertainty and provenance and measures task-oriented metrics relevant to the underwriting, accumulation management and surge planning.

2.6 Conclusion

In summary, the existing literature supports for a practical and cohesive approach to insurer decision making strategy, which includes employing GLM as an interpretable baseline with gradient boosted models (XGBoost) as a main to enhance accuracy. The inclusion of spatial diagnostics can help to gain insights into clustering and neighbourhood effects and clustering.

Moreover, by linking model outputs to county level map centric perspective for the users to filter and manipulate with straightforward scenarios. Furthermore, the dashboard will reveal uncertainty and provenance and should be assessed using task-oriented metrics, this combination informs that the developing framework will implement reproducible pipelines by cleaning and joining UK county level climate and disaster data and insurance claims. It will also train and serve both GLM and XGBoost with Poisson and Tweedie models for claim counts and amounts. The map first interface will let users filter by county, disaster or time period and compare results across to turn them into fast and usable insights.

3. System Requirements and Analysis

3.1 Introduction

This chapter converts the aims of this project into precise, ready to build specifications. High level requirements are defined and prioritised using MoSCoW method (Must have, Should have, Could have and Won't have) to ensure that both functional and non-functional requirements remain clear before design and implementation stages. This will result in a concise guideline for the development process and subsequent testing.

The chapter is organised accordingly. Section 3.2 explains the process of gathering and prioritising requirements, Section 3.3 provides detailed account of the datasets and the refinement procedures needed to provide county level analysis which included natural disasters and climate features and the simulated insurance claims panel. Section 3.4 outlines the Legal, Social and Ethical considerations that can influence the data usage and system implementation.

3.2 High-level Functionalities

3.2.1 MoSCoW Prioritising High Level Requirements

No.	High-Level Requirement	Priority
1	The system must load and validate the required datasets (both raw and processed) and normalize county keys to ensure consistent joins.	Must Have
2	The system must feature a Dashboard Overview that includes interactive Plotly graphs and summary tables pertaining to the UK Climate, Insurance Claims and Natural Disasters.	Must Have
3	The system must feature a Data Explorer equipped with year range filters, county selectors, leaflet choropleths, interactive plotly graphs and summary tables.	Must Have
4	The system must feature a Spatial Statistics that computes the Global Moran's I and render a Moran scatterplot for a selected metric and year.	Must Have
5	The system must feature Predictive Insights that presets county level backtests and forecasts for claim counts and amounts derived from the GLM and XGBoost.	Must Have
6	The system must feature an About & Download that presents content and facilitates the downloading of prepares datasets and outputs.	Must Have

7	The system must incorporate scripts to construct the monthly panel, train, and score models, and generate predictions and metrics used by the application.	Must Have
8	The system should implement scenario controls (with presets of baseline, wet windy and extreme and sliders for precipitation and wind gust) exclusively for future forecast months.	Should Have
9	The system should cache, and re-use trained model objects from the data and models to prevent retraining during the application use.	Should Have
10	The system should validate availability at runtime and display user friendly messages if anticipated inputs are absent.	Should Have
11	The system should ensure readable units and labels and maintain consistent styling across maps, charts, and tables.	Should Have
12	The system should support a disaster uplift option in future feature building to enhance stress event frequency and impacts.	Should Have
13	The system could introduce residual based choropleths and additional diagnostics to assess model fit and spatial patterns.	Could Have
14	The system could compute local clustering lick LISA and Getis-Ord Gi* and produce mapped hot and cold spot choropleths	Could Have
15	The system won't alter raw sources datasets directly as it will read and generate processed outputs.	Won't Have
16	The system won't retrain models interactively within the UI	Won't Have
17	The system won't implement stress adjustments for historical backtesting periods.	Won't Have
18	The system won't process real time feeds or arrange background data updates within the application.	Won't Have

3.2.2 Determining and MoSCoW Prioritising the Functional Requirements

User	Requirements	Priority
End-User	The end user must be able to select the Dashboard Overview tab to view the interactive plotly charts and summary tables by selecting a dataset (UK Climate, Insurance Claims and Natural Disasters) and specific time frame.	Must Have
End-User	The end user must be able to select the Data Explorer tab to view the interactive leaflet choropleths, plotly charts and summary tables by selecting a dataset (County Climate, Insurance Claims and Natural Disasters), specific county and time frame.	Must Have
End-User	The end users must be able to select the Spatial Statistics tab to compute the Global Moran's I and view a Moran scatterplot for a chosen metric and year.	Must Have
End-User	The end user must be able to select the Predictive Insights tab to view actuals, backtests and forecasts for claim counts and amounts from the GLM and XGBoost.	Must Have
End-User	The end user must be able to select the About & Download tab to understand the app and download prepared datasets or outputs.	Must Have

Project Developer	The project developer must prepare the folder structure and place raw files so the system can utilise them.	Must Have
Project Developer	The project developer must execute scripts to construct and place raw files so the system can utilise them.	Must Have
Project Developer	The project developer must arrange the processed data and model data to ensure that the applications remain operational without the need to runtime retraining.	Must Have
End-User	The end user should be able to adjust the scenario presets and sliders to stress only forecast months and immediately see updates forecasts.	Should Have
End-User	The end user should receive explicit notifications when expected files are missing or inputs are invalid.	Should Have
Project Developer	The project developer should cache trained model for future use and prevent unnecessary computation during the applications operation.	Should Have
Project Developer	The project developer should implement run time checks to insure that end users see friendly error messages instead of crashes	Should Have
Project Developer	The project developer should implement disaster uplift options.	Should Have
End-User	The end user could toggle residual based choropleths and additional model fit diagnostics to evaluate error patterns.	Could Have
End-User	The end user could implement local clustering like LISA or Getis-Ord Gi* and visualise mapped hot and cold spots.	Could Have
Project Developer	The project developer could schedule offline tasks to update processed datasets prior to deployment.	Could Have
End-User	The end user won't be able to retrain models from the UI	Won't Have
End-User	The end user won't be able to modify historical backtest results using stress controls.	Won't Have
Project Developer	The project developer won't modify raw source datasets directly.	Won't Have
Project Developer	The project developer won't set up the application to process real time data.	Won't Have
Project Developer	The project developer won't set up the application to retrain machine learning models interactively within the UI.	Won't Have

3.2.3 Determining the Non-Functional Requirements

In addition to the functional requirements, the system also must meet and fulfil the non-functional requirements to guarantee its usable and efficient for the end users. These requirements will define essential qualities such as usability, reliability, portability, scalability, and security which are crucial for insurers, analysis, and policymakers to trust and adopt the dashboard in real-world decision-making scenarios.

3.2.3.1 Usable

- The user interface must be intuitive, with consistent styling, labelling and layout across all tabs (Dashboard Overview, Data Explorer, Spatial Statistics, Predictive Insights, and About & Download)

- Interactive visualisations such as charts and maps must deliver quick feedback when changing filters and parameters.

- Output and datasets should be downloadable by the end users in standard format (CSV or PNG) without the requirement of technical knowledge.

- The help text, tooltips and the About section must articulate system functionality in layman's terms.

3.2.3.2 Reliable

- The system must manage missing or incomplete datasets effectively, providing informative error messages rather than failing without notice.

- All data processing scripts must output processed data in a reproducible and predetermined manner.

- Predictive models must yield non-negative predictions and ensure validation and clamping.

- Back-end modules must demonstrate resilience against unexpected unput formats.

3.2.3.3 Portable

- The application must be accessible on any platform without a need for a particular operating system.

- Data paths must be related to avoid platform specific hardcoding.

- The system must utilise open-source libraries to guarantee portability without licensing constraints.

3.2.3.4 Scalable

- The system must be able to support the increasing amounts of county level climate and claim data which can amount to millions of rows while also maintaining the system interactivity.

- Data aggregation functions (daily → monthly → yearly) must facilitate scalable summerisation across numerous counties.

- Processed datasets must be cached to minimise repeated computations which enhance performance.

- The modular architecture should allow the addition of new datasets and models with minimal disruptions.

3.2.3.5 Secure

- The system must not allow raw datasets to be modified and only read them and write outputs to separate folders.
- The system must not permit arbitrary file uploads or code execution by end users.
- Downloads must be restricted to approved datasets and outputs to ensure that access to server-side files outside the project scope is prevented.
- Sensitive data (if integrated with actual claims datasets in the future) must be anonymized and aggregated at the county level to safeguard privacy.

3.3 Dataset Sources & Refinement

The development of the dashboard requires to integrate various heterogeneous datasets to document natural disasters, climate variables and insurance claim behaviour and pattern at the county level for the UK. Since the original dataset came from a diverse source with different structures and degrees of granularity, significant refinement was required to guarantee consistency and suitability for building the machine learning models. This section will outline the sources and preprocessing steps that applied to each of the datasets.

3.3.1 Natural Disaster Dataset

The Natural Disaster Dataset is sourced from (*EM-DAT - The International Disaster Database*, n.d.) which offers a comprehensive event level attribute that encompasses hazard classification, location, dates, impacts, and financial losses. For the purposes of this project, the dataset has been filtered and aligned to county geometries and aggregated down to an analysis ready scheme that corresponds to the monthly modelling granularity that is utilised throughout the system.

Refined Fields Refined:

‘Start Date’, ‘Disaster Type’, ‘Disaster Subtype’, ‘Event Name’, ‘Total Deaths’, Total Affected’, Insured Damage (‘000 US$), ‘Insured Damage, Adjusted (‘000 US$)’, ‘Total Damage (‘000 US$)’, ‘Total Damage, Adjusted (‘000 US$)’ and ‘Admin Units’

Refinement Steps:

- The ‘Start Date’ is constructed from ‘Start Year’, ‘Start Month’ and ‘Start Day’ (setting the missing ‘Start Day’ to 1), and a monthly key “ym = floor_date(`Start Date`, "month")” is derived for aggregation in helper/build_panel.R.
- ‘Admin Units’ are parsed, names are normalized according to the rules set for the project (lowercase, trim, & → and, remove non-alphanumeric), and a join is performed with the counties in the gb.json (“normalize_name()” and joins in helpers/load_data.R).

- Rows that correspond to the valid counties within the geometry domain of gb.json and the records that cannot be reliably mapped are discarded.

- Event rows are aggregated into county-month features utilised by the panel and UI (for instance, event counts and sums of impact like deaths and damages).

Justification:

- By utilising the 'Start Date' for generating ym ensures temporal stability with monthly climate and claims data which is essential to facilitate logical integrations and model features on different tab/pages and models

- Assigning counties from 'Admin Units' and normalizing names to a single join key mitigates the risk of geocoding errors and ensures compatibility with the county polygons in gb.json.

- Maintaining a concise set of hazard and impact fields retains the fundamental exploratory signals like type, severity and damages while simultaneously managing memory and load times in Shiny and enhancing reliability and responsiveness.

- Preserving both nominal and adjusted currency columns retains provenance and facilitates analysis favour adjusted values for cross year comparison without sacrificing access to the raw data.

- Omitting unmapped or out of scope events avoid artificial aggregates and county-month summaries and raw counts are intuitive for stakeholders and support spatial analysis for Moran's I.

3.3.2 County Climate Dataset

The County Climate Dataset is sourced from (*Weather Query Builder | Visual Crossing*, n.d.) which offers daily observations categorized by county name per timestamp encompassing a broad range of meteorological variables. For the purposes of this project, the data is cleared to a uniform schema, county names are normalized to the join key throughout the system.

Refined Fields Retained:

'name', 'datetime', 'tempmax', 'tempmin', 'temp', 'dew', 'humidity', 'precip', 'precipcover', 'snow', 'snowdepth', 'windgust', 'windspeed', 'windir', 'sealevelpressure', 'cloudcover', 'visibility', 'solarradiation', 'solarenergy', 'unindex' and 'icon'.

Refinement Steps:

- Convert and standardize the datetime into a proper Date Type while forcing meteorological columns to numeric as required.

- Normalised the name to county join key utilised with gb.json using the project rule set of lowercase, trim replace & with 'and' and there remove non-alphanumeric ("normalize_name()" and joins in helpers/load_data.R).

- Eliminate raw field like feelslike variants, precipprob, preciptype, severerisk, sunrise, sunset, moonphase, conditions, description, stations to minimise noise and reduce memory usage.

- Conduct basic quality checks for unlikely values and isolate missing entries before aggregation in the modelling pipeline.

- This also aggregates to monthly features in the panel builder with appropriate variables statistics, such as sums for precip and snow, means for temperature, humidity, wind speed and pressure, a suitable approach for precipcover as average coverage and a circular mean for windir to avoid and prevent wrap around bias (helper/build_panel.R, and functions is helpers/data_explorer_helpers.R)

Justification:

- To ensure that a cleaned daily row can be accurately integrated into the monthly key ym, temporal and spatial alignment should be achieved so that correct connections to disasters and claims without any discrepancies.

- Maintaining a concentrated set of interpretable variables helps retain the signal for insurance related mechanism while enhancing performance and responsiveness in the Shiny App.

- Normalizing the county names to a single join down guarantees reliable joins to the county polygons and avoids unrecognized omissions in the maps and spatial analysis.

- Utilising aggregation rules that are suitable for variables such as circular mean for wind direction, produces interpretable monthly summaries that can allow for both exploration and modelling.

3.3.3 UK Climate Dataset

The UK Climate Dataset is sourced from (*Weather Query Builder | Visual Crossing*, n.d.) which offers daily observations on a national scale, which reflects a wider climate trend. The dataset is cleaned to a uniform scheme.

Refined Fields Retained:

‘name’, ‘datetime’, ‘tempmax’, ‘tempmin’, ‘temp’, ‘dew’, ‘humidity’, ‘precip’, ‘precipcover’, ‘snow’, ‘snowdepth’, ‘windgust’, ‘windspeed’, ‘windir’, ‘sealevelpressure’, ‘cloudcover’, ‘visibility’, ‘solarradiation’, ‘solarenergy’, ‘unindex’ and ‘icon’.

Refinement Steps:

- The file is loaded with datetime and converted to Date format and the name label has been normalised. Eliminate any raw fields such as feelslike variants, precipprob, preciptype, severerisk, sunrise, sunset, moonphase, conditions, descriptions, and stations.

- The data set is used for the Dashboard Overview only so basic to minimal arithmetic sum and mean are used for the month or year integrated into the ml_monthly_panel.csv.

Justification:

- Maintaining this dataset separated from the panel helps maintain an exceptionally clean separation between the contextual signals and the predictive features at the county level. The project's focus is on the county level analysis, therefore the national level helps to see the overall conditions and dive deeper into the analysis.

- By using minimal arithmetic sum and mean for month and year visualising simple logic and fast computation while also avoiding any additional computation which is not required by the current modelling pipeline.

3.3.4 Simulated Insurance Claims Dataset

The Simulated Insurance Claims Dataset was generated from the refined County Climate Dataset, and the refined Natural Disaster Dataset. These datasets were merged using a normalised county key and aligned with the corresponding dates. The code base for this was written in the Python Notebook and used for building the panel.

Generate Fields:

'claim_id', 'county', 'claim_date', 'disaster_type', 'event_name', 'description', 'claim_amount', 'disaster_year', 'insured_flag' and 'severity_label'.

How the Claims were Simulated:

- The climate data is parsed with a parameter "dayfirst= True", and then normalised "name_norm" is generated to match counties. The disaster is then parsed from the 'Start Date' into "event_date".

- 'Admin Units' are parsed in a JSON like format to extract the adm2_name, and adm1_name. The counties are converted to lowercase to match.

- A 29-day period is implemented for each disaster and affected county which spans from 7 days before to 21 days following the event_date to capture lead and lag effects.

- A day is classified as an extreme if any of the following conditions are met: precip > 15, precipcover > 50, snow > 5, snowdepth > 2, windgust > 40, windspeed >30 and a lower, secondary threshold (precip >5 or windgust > 25 or snowdepth > 1) are also considered valid.

- The 'severity_label' is designated as High if it is two or higher of the above triggers occur on a given day otherwise it is classified as a Low.

- For each extreme day, a random integer representing the number of claims is uniformly selected from the range of 1 to 20.

- Each claim is then assigned a positive amount drawn uniformly from a range of 2.000 USD to 100,000 USD rounded to two decimal places.

- Each entry is therefore recorded with the claim_id ("c" + 12-character hexadecimal string), county (from 'Admin Units'), 'claim_date' (the extreme day), 'disaster_type', 'event_name', 'description' (that days field from the Natural Disaster Dataset), 'disaster_year' (from the 'claim_date'), 'insured_flag' (drawn with 70% Yes and 30% No) and 'severity_label' (based om Highs or Lows trigger).

- The resulting dataset is sorted by 'claim_date' and 'county' (No random seed is set, resulting in varying outputs across different runs).

Justification:

- Generating claims solely for counties explicitly mentioned in each event's 'Admin Units' ties claims to documented impact zones, thereby avoiding misleading spatial spillover.

- Implementing the -7- and +21-days window around each 'event_date' captures plausible pre-event and post event aftermath when claims are most likely to rise.

- The claims are only triggered on days that breach the clear thresholds or 'Icon' validated lower thresholds to ensure the severity, no noise and drives events.

- Normalizing the county names guarantees matched and prevents silent joint failures that could compromise counts and amounts.

- Generating 1-20 claims per extreme day results in a realistic surge volume while also preventing runaway magnitudes that could overshadow the monthly aggregates.

- Sampling amounts from 2,000 to 100,000 USD produces reasonable severities, avoids negatives and zero and limits outliers that could jeopardise the model.

- Keeping 'disaster_type', 'event_name', and 'icon' can come handy when developing stress test model.

3.3.5 Monthly Panel Dataset

The Monthly Panel Dataset is constructed from the Simulated Insurance Claims Dataset, Count Climate Dataset and Natural Disasters Dataset.

Resulted Schema:

'county', 'ym', 'claim_count', 'claim_amount', 'precip_sum', 'snow_sum', 'windspeed_mean', 'windgust_mean', 'humidity_mean', 'tempmax_mean', 'tempmin_mean', 'lag1_precip_sum', 'lag1_windgust_mean', 'disaster_count', 'month_num'.

Construction Process:

- Claim records are parsed by date, sorted by monthly key ('ym') and grouping by county and month to generate two targets, the number of claims ('claim_count') and the total claim amount ('claim_amount'). The names of the counties are normalised to ensure they match with the mapping geometry and other datasets.
- These daily county observations are converted into monthly features for each county and month by employing statistics that are suitable for the variables such as total for precipitation and snow, means for temperature, humidity, windspeed and windgust. Following the monthly aggregation, short term persistence is incorporating one-month lags ('lag1_precip_sum', 'lag1_windgust_mean').

- The disaster events are analysed by month (ym) the list of 'Admin Units' is expanded to include each individual counties, name are than normalised, events are counted per county and month to get disaster count.

- A comprehensive grid representing counties and months is established covering the entire study period by the three datasets. Subsequently, the monthly targets (claims) and features (climate and disasters) are incorporated into the grid by left join based on (county, ym) therefore maintain a consistent panel even when individual datasets are sparsed.

- Missing values resulted due to joining are handled by setting them to zero when suitable. The 'month_num' (1-12) is calculated to account for seasonality effects, and the finished panel is saved.

Justification:

- By employing a standardised monthly key, it guarantees arrangement between claims, climate and disaster drives which is essential for accurate modelling and analysis.

- Developing a comprehensive county-month lattice explicitly reveals zeros and differentiates time absence of events from the missing data therefore enhancing both interpretability and model training.

- Utilising variable specific aggregation maintains and retains physical meaning of the measures while minimizing noise.

- By having a simple one-month lags which captures short term dependencies of surge behaviour without unnecessarily complicating the feature space, therefore ensuring the panel remains stable and reproducible for the subsequent prediction panel.

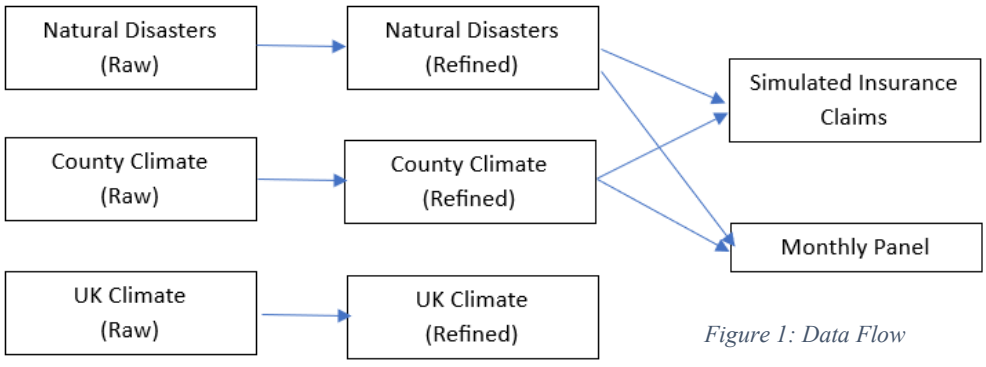


Figure 1: Data Flow

3.4 Legal, Social and Ethical Issues

3.4.1 Legal

The project will use information from natural disasters and insurance claims, which necessitates careful consideration of the UK Data Protection Act of 2018 (DPA) and the General Data Protection Regulation (GDPR). These legal frameworks demand that any personal or sensitive data must either be anonymized or pseudonymized to safeguard the identities and privacy of individuals. To maintain compliance, the project will use an anonymized dataset or simulated data that omits any personally identifiable information.

The utilization of datasets that may be obtained from public or licensed platforms will require strict meticulous consideration and adherence to copyright and licensing regulations. For instance, any dataset that is available under the Open Government License (OGL), which allows free usage if proper attribution is provided. Adhering to these attribute requirements and upholding all intellectual property rights is essential in ensuring the project's legal integrity.

Considering the interactive characteristics of the dashboard and the possibility of handling sensitive information, the project will comply with recognized security protocols. As such, when the dashboard is deployed online, mechanisms to combat unauthorized access, injection flaws and insecure data access controls will be required to be established to avert unauthorized use. By following these standards, the system will satisfy most of the legal obligations but also foster confidence in its design and execution.

3.4.2 Social

The project will ensure that it meets the societal norms, notably in terms of accessibility and risk of misunderstanding and misinterpreting any potential outcomes. Accessibility is a fundamental component in this project, since it will be used by a wide range of users which includes individuals with disabilities. The project will comply with Web Content Accessibility Guidelines (WCAG), which provides a framework for guaranteeing digital accessibility. This involves incorporating features such as high contrast settings, reader compatibility and intuitive navigation to support users with visual or motor challenges. By prioritizing accessibility, the project not only adheres to the Disability Discrimination Act of 1995 but also embraces the wider notion of inclusiveness in technology.

If users lack a comprehensive understanding of the tool's limitations, the machine learning model and visualization may cause misunderstanding. For instance, predictors are fundamentally probabilistic and should not serve as the key choice when determining insurance premiums or distributions of resources. To address this concern, the system will incorporate explicit warnings about the machine learning model assumptions, data constraints and biases. This will keep users well informed and help minimize unexpected outcomes, such as excessive dependence on predictions in the decision-making process.

Another social issue is the capability of the tool to expose regional disparities in disaster planning and response. By pinpointing regions that experience greater claim volumes or expenses, the system can illuminate areas that are neglected thereby encouraging more equal support allocation. Nevertheless, it is important to exercise caution to prevent labelling certain regions as high risks in a manner that might result in unjust treatment, such as higher insurance premiums

for inhabitants of those areas. The application needs to prioritize to communicate the finding of the tool to promote fairness and equity.

3.4.3 Ethical

Predictive models are fundamentally based on historical data, which can often harbour intrinsic biases. For instance, insurance claims related to disaster may be skewed towards areas with better insurance coverage, thereby neglecting regions that are underserved or underinsured. Failure in recognizing and addressing these biases could result in predictions that perpetuate disparities or give incorrect information. To counteract this issue, the project will conduct a comprehensive study of the dataset to identify any potential biases and employ tactics such as balanced sampling and feature normalization to promote fairness in model result. Additionally, regular review of the model's predictions will aid in detecting and correcting any unwanted biases.

Transparency and clarity play a vital role in fostering user trust and encouraging ethical systems use. The machine learning models inherently operate on probabilistic principles and are based on assumptions that the users must understand in order to interpret the outcomes responsibly. Furthermore, the dashboard will feature comprehensive documentation and warning that describe how the process of prediction is made, constraints of the data and the extent of the insights presented.

The system will emphasize non-discriminatory methods by refraining from utilizing demographic or socio-economic factors that might result in biased treatment of certain groups and areas. For instance, forecast would exclude data elements that might contribute to discriminatory pricing for vulnerable people. Furthermore, the tool will be structured to promote fair resource distribution and disaster preparation as well as focusing on supporting marginalized regions instead of labelling them as high-risk.

4. Design Development

4.1 System Architecture

The NatInsureR system is structured with modular architecture that promotes a transparent separation of data, analytics, and interface layers.

The data directory has the repository of both refined datasets (county climate, UK climate, natural disasters, and simulated insurance claims), processed machine learning input/output (monthly panels, metrics, and forecasts), geospatial assets (gb.json), and trained models (GLM and XGBoost on counts and amounts).

Within the EDA folder is the repository for the scripts and notebooks that conduct exploratory data analysis on each dataset

The helpers directory offers a reusable helper functions that conduct modelling, panel building, spatial analysis, evaluation, downloads, and data loading.

The predictive models run organised within the helpers/models subdirectory that is specified in terms of extensibility.

The server and ui directories encapsulate the Shiny back-end and front-end logic respectively, with scalability and maintainability controlled through separate modular pages (it has the logic and ui for the following pages, Dashboard Overview, Data Explorer, Spatial Statistics, Predictive Insights and About & Download).

Supporting documents are stored in the docs folder, while static assets such as the apps logo reside in www.

At the top of the root level, app.R integrates all the modules, datasets, and models into an end-to-end R Shiny dashboard that allows for an interactive exploration, forecasting, and support for making informed decisions.

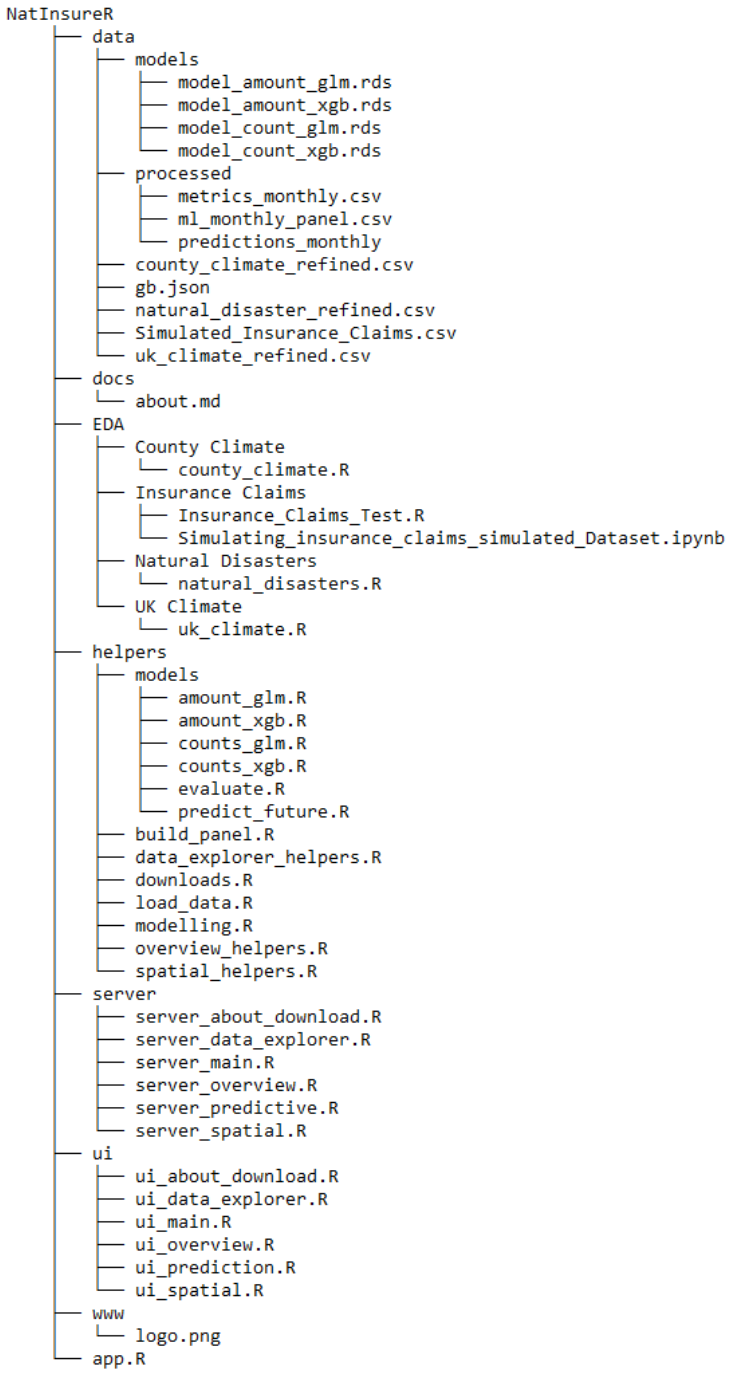


Figure 2 : System Architecture

4.2 Dashboard Design

In this section the initial design of the dashboard will be implemented by using wireframe to get a low fidelity wireframe and a mid-fidelity wireframe.

4.2.1 Low Fidelity

4.2.1.1 Dashboard Overview

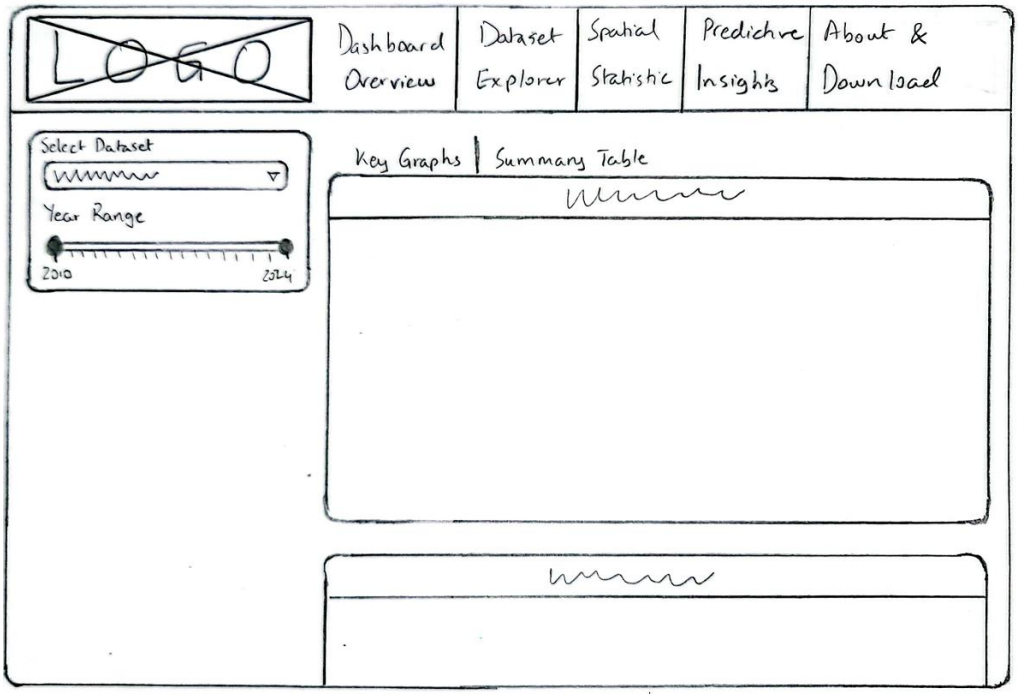


Figure 3 : Low Fidelity Dashboard Overview

4.2.1.2 Data Explorer

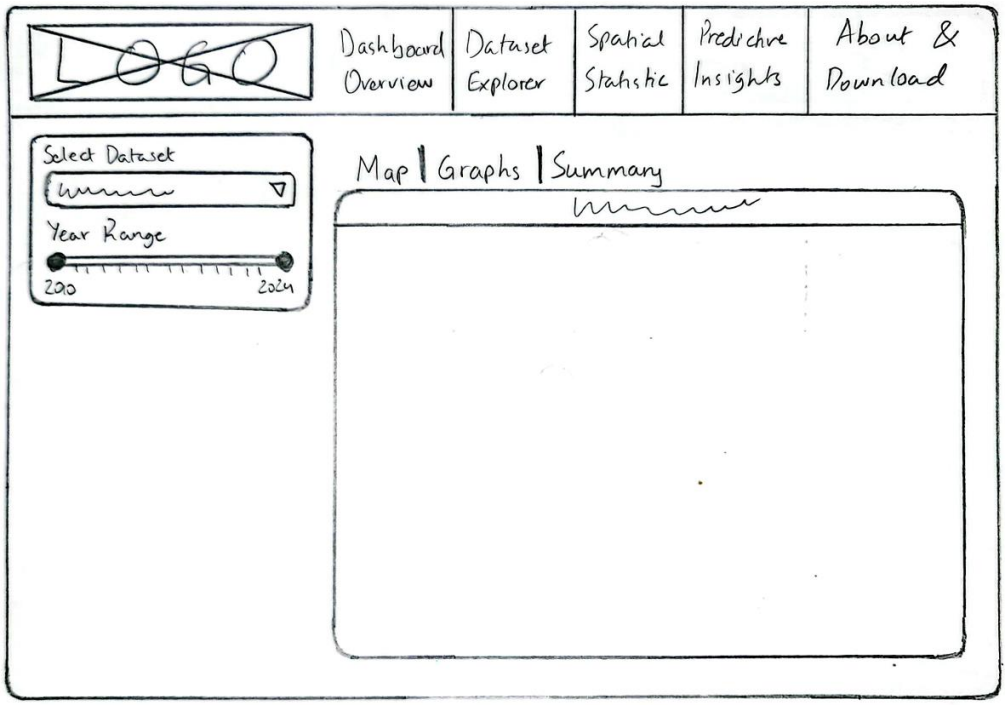


Figure 4 : Low Fidelity Data Explorer

4.2.1.3 Spatial Statistics

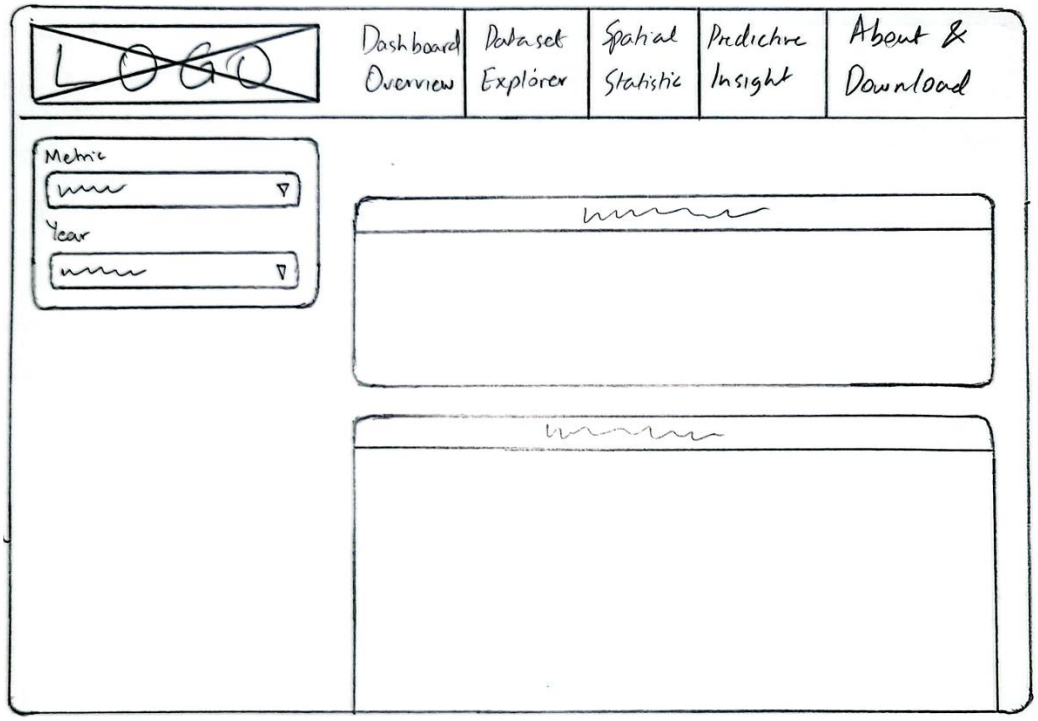


Figure 5 : Low Fidelity Spatial Statistics

4.2.1.4 Predictive Insights

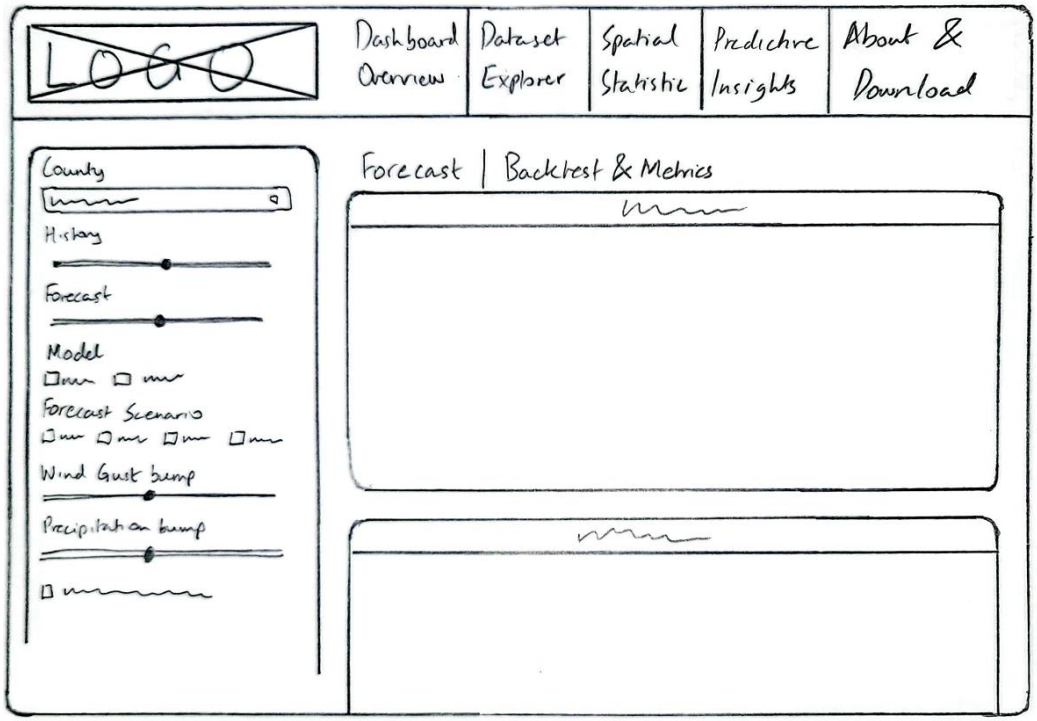


Figure 6 : Low Fidelity Predictive Insights

4.2.1.5 About & Download

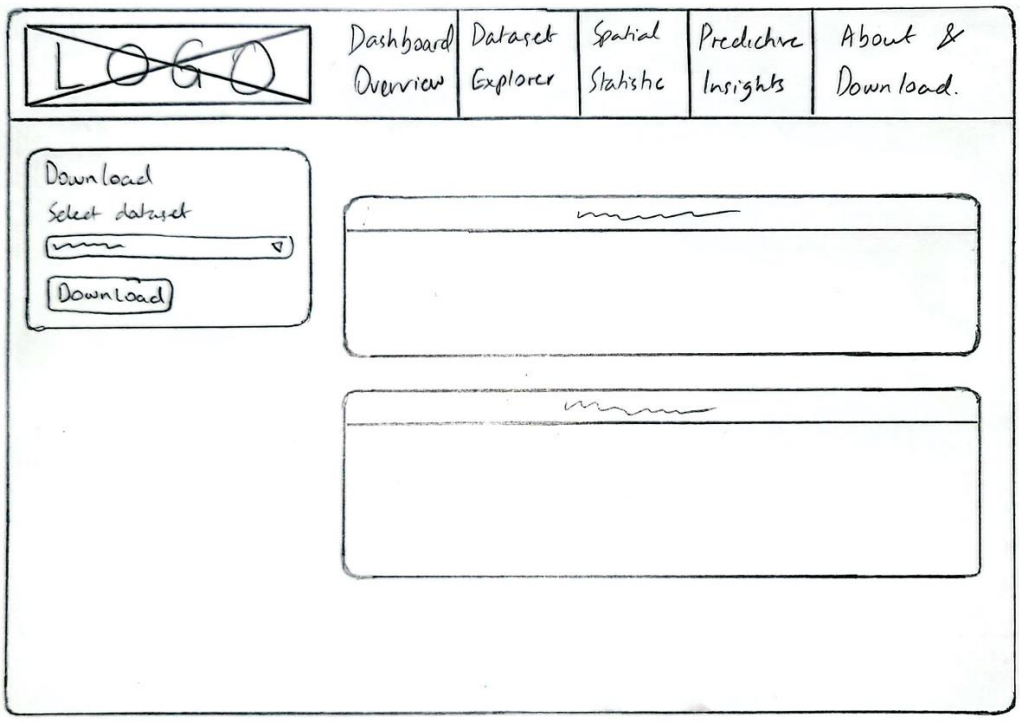


Figure 7 : Low Fidelity About & Download

4.2.2 Mid Fidelity

4.2.2.1 Dashboard Overview

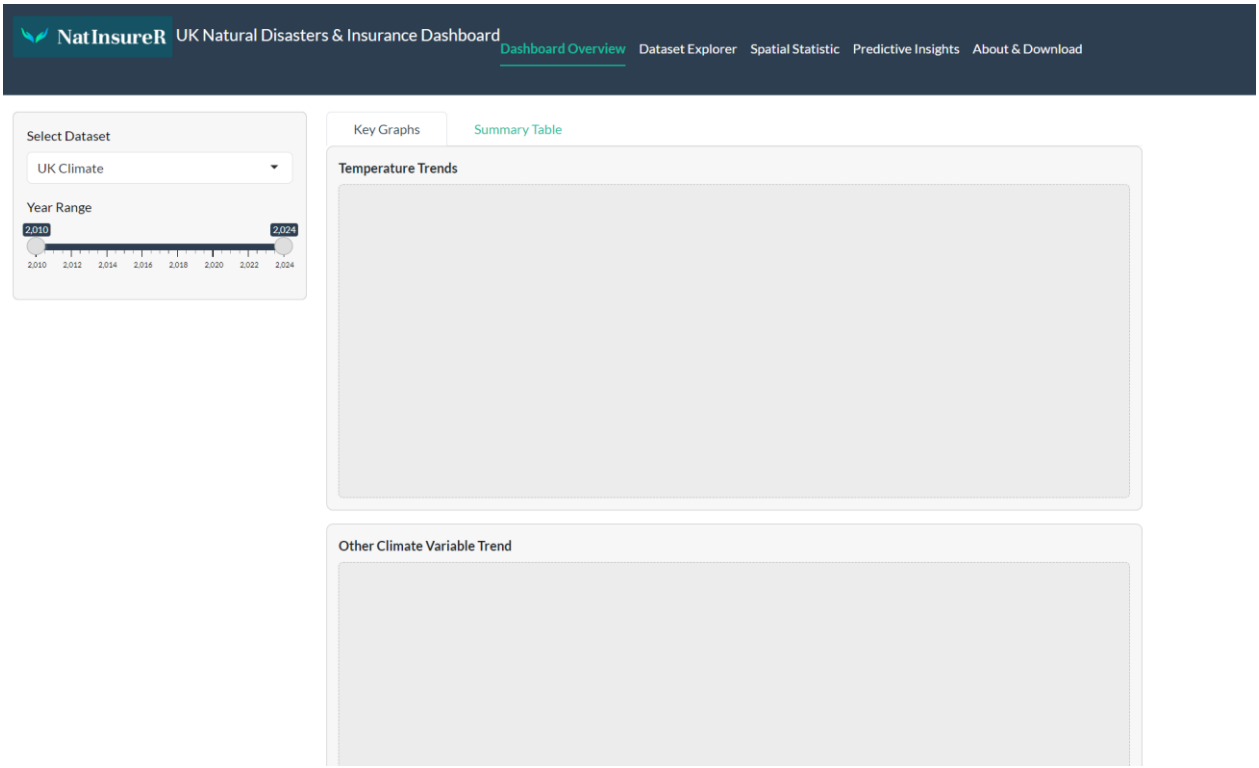


Figure 8 : Mid Fidelity Dashboard Overview

4.2.2.2 Data Explorer

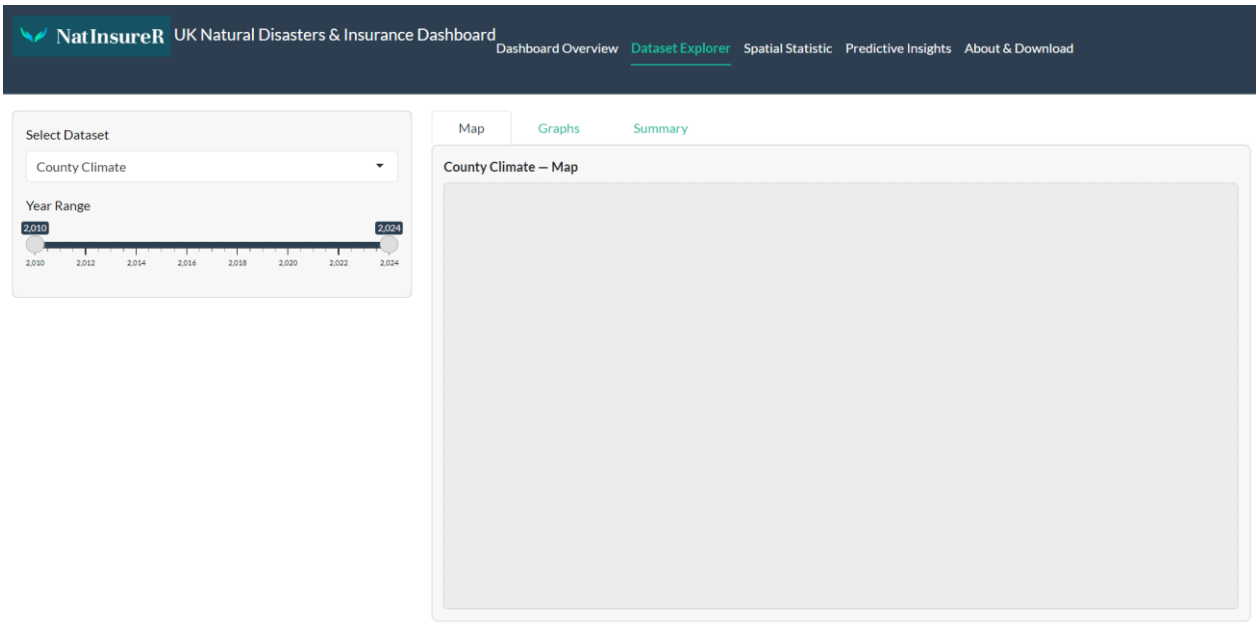


Figure 9 : Mid Fidelity Data Explorer

4.2.2.3 Spatial Statistics

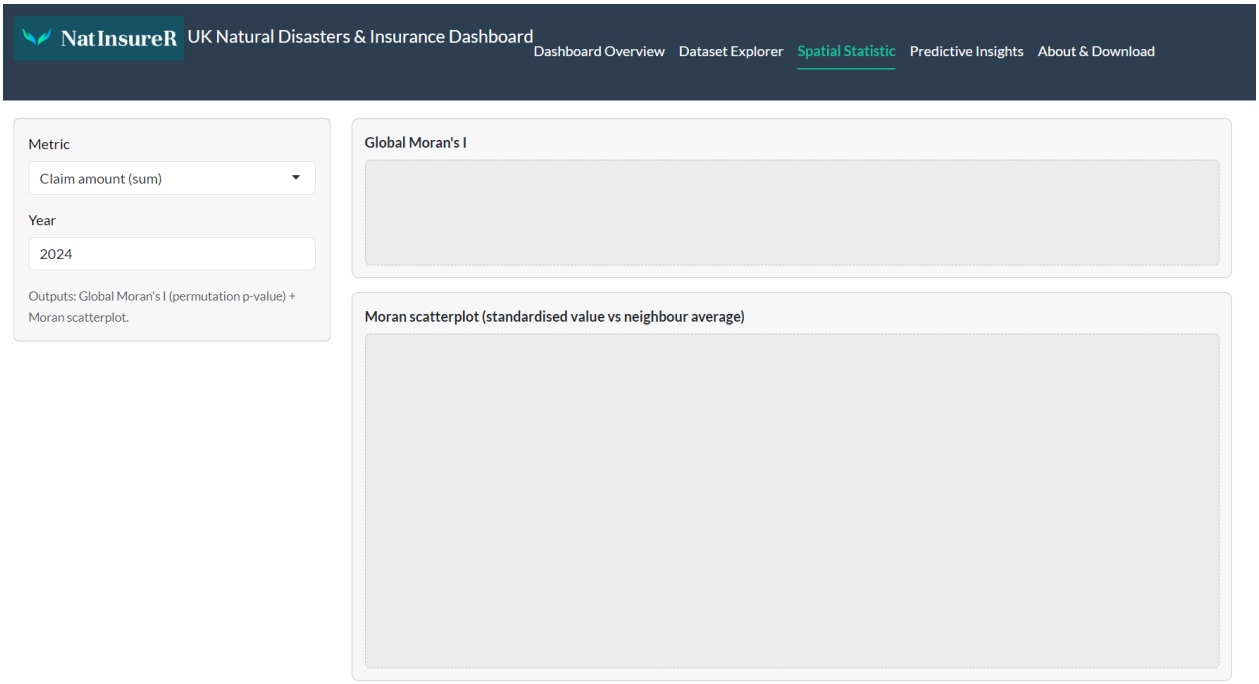


Figure 10 : Mid Fidelity Spatial Statistics

4.2.2.4 *Predictive Insights*

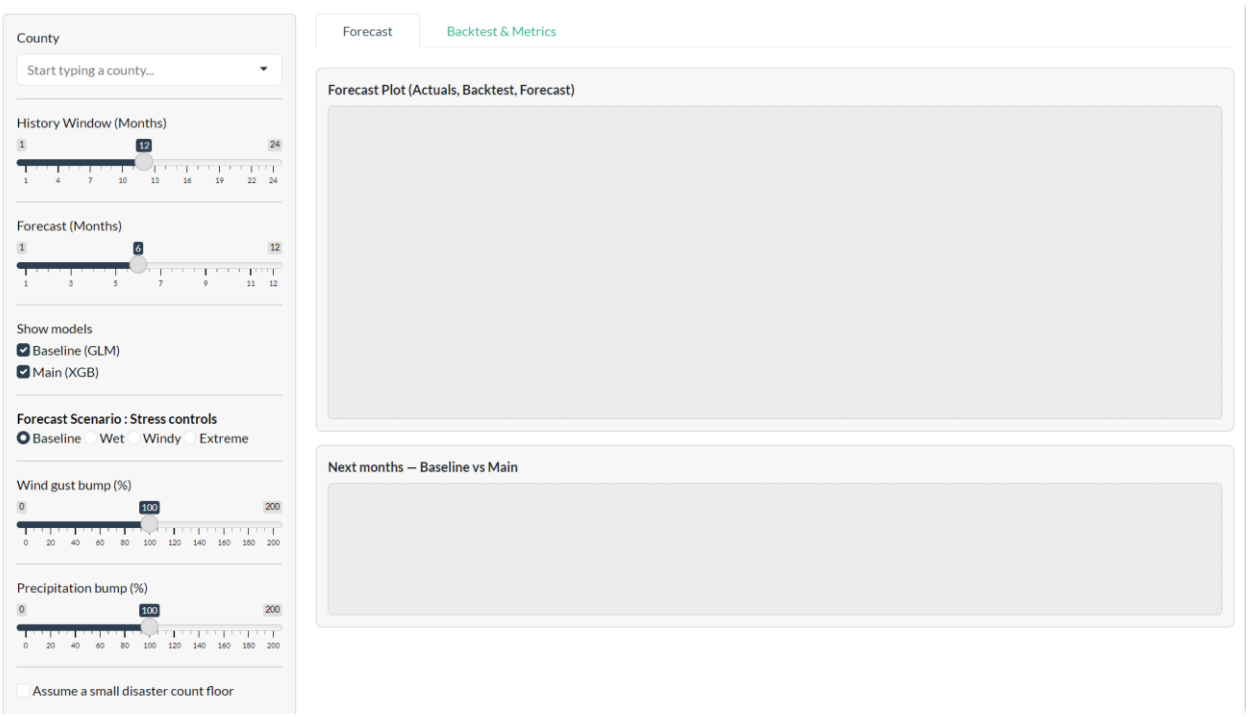


Figure 11 : Mid Fidelity Predictive Insights

4.2.2.5 *About & Download*

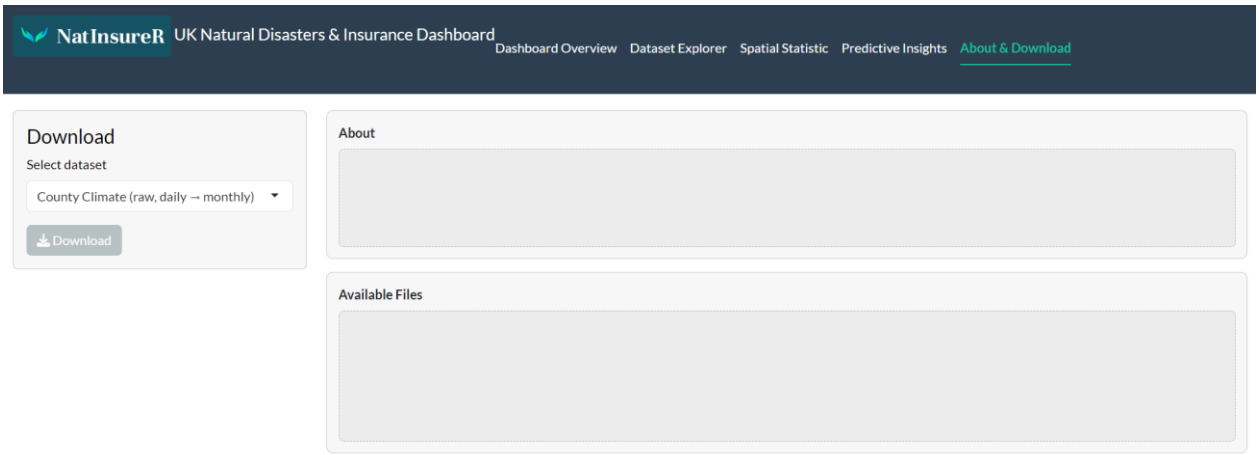


Figure 12 : Mid Fidelity About & Download

5. Implementation

This chapter explains the process through which the proposed system was developed and transformed into a functional prototype. While Chapter 4 outlines the general architecture and design decisions, this section emphasises the concrete actions undertaken to implementing those ideas using R and Shiny. Each individual part of the system is laid out regarding its function, the development process adhered to, and the strategies employed to manage errors or unforeseen circumstances. The complete source code is included in the appendix, whereas this chapter offers detailed explanations and selected outputs to illustrate the system's functionality.

This chapter commences by outlining the background functionalities that support the entire application, which encompasses development, loading the dataset and creating the monthly panel. Subsequently, it navigates through the distinct tabs of the application in the sequence they are presented in the interface, Dashboard Overview, Data Explorer, Spatial Statistics, Predictive Insights and ultimately the About & Download section. This will help illustrate the modular architecture of the system and guarantee that each feature is explained in detail in relation to its function within the comprehensive application.

5.1 Background Functionalities

5.1.1 Deploying App

The application is launched via a single-entry script which is the "app.R". It is used to integrate all the features of the system together. The script initially imports the necessary libraries for user interface design, data manipulation, visualisation, spatial analysis, and modelling. Subsequently, it sources auxiliary functions to prepare and analyse the datasets, along with the UI and server modules for each corresponding tab.

The user interface is structured within district UI modules and has been integrated into a unified layout by the file "ui_main.R". The server logic has also a similar modular structure with each tab being maintained by a separate server file and coordinated via "server_main.R". When executing the function (shinyApp(ui = ui_main, server = server_main)) launched the system. This guarantees that both the user interface and server features function cohesively as a complete application.

Furthermore, this modular approach enhances maintainability because each tab can be modified independently and is consistent with the standard practices for bigger Shiny Applications.

5.1.2 Load Dataset

Objective

The aim of this feature is to establish a unified and reliable source of data for all modules of the system. It guarantees that raw datasets, processed datasets, and geospatial resources are loaded at a single time and made available throughout the entire system. Additionally, another separate spatial loader has been developed to facilitate the Spatial Statistics tab which integrates shapes, the monthly panel, and the prediction data.

Process

The loading process is initiated with the spatial only loader, which imports the county polygons from the GeoJSON file, standardizing the county names and verifies that the monthly panel contains essential columns such as claim_count and claim_amount. Additionally, it also incorporates forecasts to facilitate residual computations in the Spatial Statistics tab.

On the other hand, the project wide loader gathers all additional resources. When accessible, raw datasets such as UK Climate, County Climate, Insurance Claims and Natural Disasters are utilised in the Dashboard Overview and Data Explorer tabs. Furthermore, the loader reads the processed outputs such as monthly panel, predictions and the evaluation metrics which are crucial for predictive modelling. The county geometrics undergo a robust naming routine to guarantee that joins stay stable and consistent across various shapefile attributes. Additionally, a download catalog is created, which includes file names, paths, and sizes for reference in the About & Download tab.

Error Handling

Each dataset is processed within “tryCatch” blocks. In cases where a file is absent or unreadable, the loader sets it to NULL therefore allowing the system to present a clear warning instead of failing. Whereas the spatial loader verifies the check for necessary columns and stops the execution with a diagnostic message if any are missing, therefore preventing misleading outcomes in spatial analysis.

Alternative Implementation Option

One alternative option could have involved loading the datasets as needed within each tab. However, this approach was rejected because of the potential for increased latency from repeated reads which would lead to inconsistencies between modules. Another possibility was to use cache, which would involve storing loaded datasets into memory across sessions. Although this method might cut load time even further, it was not given priority because the datasets at this time were manageable size, and reliability was deemed to be greater significance.

5.1.3 Building the Monthly Panel

Objective

The aim of the feature is to convert the raw datasets into a unified monthly panel that matched claims, climate characteristics and disaster events by county and time. This dataset serves as the basis for predictive modelling and offers a variety of visualisations throughout the system.

Process

The process is initiated by compiling the insurance claims into total monthly counts and amounts per county. The County Climate Dataset is converted from daily to monthly summaries which capture precipitation, wind, temperature, and humidity. Lag variables are integrated to illustrate the short-term persistence of climate conditions. The Natural Disaster Dataset is processed to retrieve county level records and is then aggregated into monthly disaster counts.

These three processes are merged to provide a comprehensive grid that displays county dara by month, guaranteeing representation for each county and time interval regardless of whether any events took place. Furthermore, any absent values have been substituted with zeros for target variables such as claims and disaster counts. The resulting dataset is then saved as ml_monthly_panel.csv and later utilised by the modelling pipeline and other tabs within the system.

Error Handling

Prior to processing, the function verifies the existence of all raw datasets. If it is unable to extract county names from the disaster dataset or if no counties are discovered, the process is terminated with an error message. Any missing values from joins are substituted with zeros to prevent discrepancies in subsequent analyses.

Alternative Implementation Option

An alternative approach would have been to generate features as needed within the predictive modelling pipeline. However, this methos was avoided to minimise runtime delays and to guarantee that all tabs use a uniform dataset, an additional expansion may be to provide a more complicated lag feature or rolling averages, although this would lead to greater complexity and longer processing times.

5.2 Dashboard Overview

Objectives

The Dashboard Overview tab offers a concise summary of the three datasets, UK Climate, Insurance Claims, and Natural Disasters. Its aim is to allow users to swiftly investigate trends, distribution, and relationships without delving into thorough research. The tab integrates interactive visualisation with summary tables to provide insights in a user-friendly manner.

Process

The user interface is developed in “ui_overview.R”, while the server logic is contained within “server_overview.R”, and supported by helper function in “overview_helper.R”. The sidebar provides users with the ability to choose the dataset and the range of years. Depending on the selections, the conditional panel displays the relevant graphs and tables.

If chosen dataset is the UK Climate, it includes temperature trends lines, line charts specific to various, histograms for selected measures, bar charts depicting precipitation and snow, scatterplots illustrating risk relationships and pie charts that represent the weather composition. These visualisations are responsive to frequency options such as daily, monthly, or yearly.

If Insurance Claims Dataset is selected, it features bar charts of claims counts and amounts that are categorised by variables such as disaster types or county, as well as yearly line plots that depict the total claims over time.

If the Natural Disaster Dataset is chosen, it highlights the line charts that represent disaster frequency, bar plots that display the distributions by type or subtype and breaks down the

variable like damage and deaths across various classes. Users also have the option to view aggregated metrics over months or years.

Alongside the visualisations, a distinct summary table displays the top records or group summaries. The UK Climate Dataset encompasses the top 100 rows by variable. For Insurance Claims and Natural Disasters datasets, the tables are organized according to the chosen variable like county or disaster types, and total are shown.

Error Handling

The server verifies that all the necessary inputs are provided before rendering the plots. For instance, if there is no temperature variable chosen, a warning appears rather than an empty chart. Filtering functions are implemented reactively to guarantee that only valid data from the specified year range is handled. In cases where datasets have missing values or empty groups, outputs are modified to avoid disrupting the interface.

Alternative Implementation Option

An alternative design idea could have been to create three independent tabs for each dataset instead of integrating them into a single module. Although this may simplify the logic, the selected method consolidates functionality and ensures a uniform user experience. Another approach would have been to pre-compute the summaries and save static visualisations, which could have reduced the required computational power, but it would have sacrificed the interactivity and flexibility.

5.3 Data Explorer

Objective

The Data Explorer is an extension of the Dashboard Overview which allows users to delve deeper into each dataset. It offers three modes of interaction, choropleth maps for analysing spatial patterns, graphs to examine the temporal and variable specific trends and the summary tables to present detailed records. This tab accommodates three datasets which are County Climate, Insurance Claims and Natural Disaster. It is intended to allow insurers and policy makers to concentrate on a particular county, metric, and time period.

Process

The user interface (ui_data_explorer.R) features a side bar to allow select datasets and year range. The panel contains three tabs which are Maps, Graphs and Summary. The server component (server_data_explorer.R) actively filters the appropriate dataset depending on the user inputs and generates the outputs utilising functions from “data_explorer_helpers.R”.

The map tab generates the choropleths maps at the county level by using leaflet. Users can choose from various climate data variables. The insurance claims map can illustrate the number of claims or total amount, whereas the natural disasters maps provide parameters such as number of deaths and unsured damages.

The graphs tab offers a customisable visualisation for each dataset. The climate data encompasses temperature trends, histograms, bar charts of precipitation and snowfall, scatterplots illustrating risk relationships and pie charts depicting weather patterns. Claims data allows for pie charts representing the proportions of insured versus uninsured as well as scatterplots correlating claims amounts with severity and coverage. The disaster data presents pie charts showing the frequency of disasters, scatterplots connecting human and financial impacts and boxplots of crucial variables.

The summary tab highlights the ranking tables for a specific variable. Climate data highlights the top 100 records, claims data compiles totals by county or type of disasters. The disasters data categorises the parameters like damage or deaths.

Error Handling

The server employs “req()” to verify the validity of the inputs and datasets before producing the charts, the year and county pickers are only supplied when data is available, therefore avoiding invalid selection. In the instance when there is no data present for the given selection, the application either display a placeholder plot (such as a neutral map or a pile chart indicating ‘No data’) or provide with an empty table instead of failing numeric conversions are conducted securely with warning suppressed when processing the fields.

Alternative Implementation Option

An alternative design choice would have been to build three separate tabs per dataset rather than combining them into one module (Similarly to the Dashboard Overview, having too many tabs in the main navigation bar would clutter and would be harder for the user to clearly distinguish between tabs). This may simplify the logic but, the chosen approach centralises the functionality and maintains a consistent workflow. Another option is to pre-compute summaries and store them as static visualisations that could reduce the computational demand but will sacrifice in terms of flexibility as well as interactivity.

5.4 Spatial Statistics

Objective

The spatial statistics module was developed in order to identify if insurance claim patterns or the model residuals indicate spatial dependence among counties in the UK. The aim was the provision of statistical evidence of clustering or dispersion, along with a visual diagnosis of spatial patterns. This function is beneficial to insurers and policymakers in gaining insight into whether claim outcomes are distributed randomly or guided through the surrounding counties to facilitate fairer prices as well as focused policy interventions.

Process

The user interface comprises two primary controls, metric and year selectors. The metric selector allows the user to study four options, which are the counts of all claims, the amounts of all claims, the counts of residuals, and the amounts of residuals. The residual metrics are particularly useful as they identify and reveal areas where models consistently under-predict or over-predict.

On the server-side, the chosen year and metric are passed into a helper function (ss_prepare_values). The function matches the polygons at the county level with the corresponding values from that year. The residual metrics are then determined by merging the monthly claims dataset with the model predictions and then computing the difference. Subsequently, the data is then aggregated as yearly totals or averages, depending on the metric.

Two outputs are produced. Initially, the Global Moran's I is calculated utilising a queen contiguity spatial weights matrix. A Permutation test involving 499 randomisations creates a p-value that determines whether the observed spatial pattern detected is statistically significant. Subsequently, a Moran scatterplot is generated, displaying standardised values from the counties along the x-axis as well as spatially lagged values (neighbour averages) along the y-axis. The quadrants in the scatterplot (High-High, Low-Low, High-Low, Low-High) allow users to observe from the scatterplot whether there are clusters or if there are any significant outliers.

Error Handling

The system verifies that at least a sufficient number of finite observations for the chosen metric and the year. In cases, if there are less than three valid counties, the module outputs an explicit message ("No data for this year/metric") rather than generating potentially misleading results. Island that lacks neighbouring areas, such as islands, are assigned zero lag, so they are guaranteed to plot correctly in scatterplots without affecting or interfering with the calculations. If the residuals or claims are missing in any of the county-years, the values are automatically disregarded.

Alternative Implementation Option

The existing implementation utilises Global Moran's I with permutation-based significance testing, whereas other techniques may take longer. Local Indicators of Spatial Association (LISA) can be employed to potentially identify and reveal hotspots and cold spots within particular counties. Similarly, Getis-Ord Gi* statistics can be utilized in the identification and pinpointing significant clusters of high or low values. Although, these will provide more detailed insights, they were omitted to preserve, maintain the scope, and avoid overwhelming the users with complex and intricate outputs.

5.5 Predictive Insights

Objective

Allow the users to select a county and then examine the recent history and backtests fits, it predicts the following N months under user-specified stress scenarios and thereafter lets the user view validation metrics for counts and amounts. The tab displays the models used which are Baseline (GLM) and Main (XGBoost), such that allows stakeholders to compare and assess their robustness against zero-heavy data and spike behavior.

Process

1. Setting up the view: The sidebar lets you pick a county (populated from server). There's also the history window which displays months of actuals and backtest, and options to set forecast horizon length, and choose which model (GLM or XGBoost) to view.

2. Pick a scenario: Users can choose from Baseline/Wet/Windy/Extreme and may optionally can tweak by adjusting granular stress sliders for windgust and precipitation bumps (%) along with a toggle for a “small disaster floor.”
3. Compiling the data: The server cleans up the data by normalising the names, fixing data types, and generating three reactive features which are monthly actuals, model forecasts, and precomputed metrics. County selections are based on the monthly panel.
4. Limit the time frame: The history is shortened to the last k months. The forecasts from the selected county are divided into Backtest and Forecast.
5. Applying stress only for future predictions: Only a single multiplier (from preset values, sliders, optional disaster lift) is multiplied with the future GLM and XGBoost predictions, whereas the backtest entries are unaffected in order not to compromise validation integrity.
6. Render outputs.
    - Forecast visualisation: it displays the Actuals (points and line), the GLM and XGBoost series, with dashed Backtest and a solid Forecast segment.
    - Next-month table: this presents the initial H forecast rows which is the counts and amounts derived from GLM and XGBoost.
    - Backtest & Metrics: A last 12-month plot of Actual vs Baseline/Main and two metrics tables consisting of Counts (MAE, WAPE, Bias ratio, Zero-hit rate) and Amounts (MAE, WAPE, Bias ratio, RMSE). There is also a detailed description to explain why XGB has typically appeared below GLM in zero-inflated scenarios.

Error Handling

• Incomplete data or columns: This helper function “.ensure_actual_amount()” is responsible creating and cleaning ‘claim_amount’. The feature builder combines climatology and ensures that the numeric drives are in place and the necessary baseline anchors are filled in (county×month → county → month → global).
• Empty data slices: If there are no rows available for a specific county or horizon, the plots short-circuits (stops) and tables return NULL, thus preventing any UI crashes.
• Stress controls: The sliders are clamped and restricted to 0–200%, where the multipliers only apply to future timestamps.
• Metric stability: Protection from division-by-zero via “eps”. Here WAPE and Bias utilise safer denominators and caps; zero-hit returns NA when there are true no zeros present.
• Model segmentation: Trains the XGBoost to ensure that both the training and validation set are not empty therefore backfilling the validation with positive months when the data is limited to zero for counts and amounts.
• Design matrix drift: Fixed factor domains (county, month) and align the sparse matrices to prevent column mismatch during prediction.

Alternative Implementation Option

• Modelling: employing the zero-inflated Tweedie-GAMs, hierarchical and Bayesian GLMs, quantile/Pinball-loss XGB, or hybrid count severity models to effectively better manage spikes and uncertainty.
• Uncertainty: Adds prediction intervals using bootstrap method, Bayesian posteriors, or conformal prediction to display fan charts.

• Backtesting: Substitute the single holdout with rolling-origin cross-validation to achieve more reliable scores.
• Scenarios: Rather than using the multiplicative bumps, simulating weather covariates from historic and ARIMA distributions or eternal seasonal forecasts to propagate these through the models to create stochastic bands of scenarios.

These enhancements would add depth and probabilistic context but were deferred in favour to maintain focus and efficient UI for fast and streamlined decision-makers.

5.6 About & Download

Objective

The About & Download module was created and designed with the concept that the module would interrelate two functions. Firstly, the module would provide users with accessible documentation describing the use and functionalities of the dashboard, thereby ensuring that the application would be clear, transparent and user-friendly. Secondly, the module would provide an organized and systematic approach of downloading essential datasets and processed outputs such as raw data which were refined, modelling panels, predicted outputs, and validation metrics. These two functions make the tab both an informative entry point and an effective way of exporting results for further in-depth analysis.

Process

The interface is separated into the sidebar and the main panel. The sidebar contains the dataset selector along with the download button. This selector is automatically populated during the apps runtime with available datasets and outputs that are registered in the system. Upon selecting the dataset and clicking the download button, the system will verify the source file and transfers the file onto the user's device.

The panel consist of two components. The first displays the “About” page, which is automatically generated from an external Markdown file (docs/about.md). Thus, the project documentation is stored outside the source code and can be maintained and updated without altering the code. Secondly, the file inventory table showcases all the downloadable datasets and outputs present. The table includes all their file paths, their existence status, and their size in human readable units. This level of transparency allows the users to confirm whether the artefacts are present prior to initiating a download.

Error Handling

The module also includes various protective measures. If the download registry does not exist or is blank, the system provides with an explicit validation message instead of leaving the selector unpopulated. Should the file “About” not exist and, the panel displays the placeholder message indicating the developers to include the “docs/about.md.” Regarding the downloads, if there is no file or file has yet to be created, the system halts the procedure and informs with an explanatory message, such as asking developers to run the modelling pipeline in order to produce some predictions. The file sizes are calculated meticulously, and any missing or damaged files appear as NA in order to prevent any potential misinterpretation.

Alternative Implementation Option

The existing design provides a clean yet dependable experience with usability and transparency in focus. Another approach would be packaging the datasets together in compressed archive like a ZIP file format for large-scale downloads, incorporating version identifiers and checksums to ensure reproducibility, or utilising cloud-based storage for managing larger files. An additional option would be the introduction of an interactive "on-demand" report generation by merging forecasts, metrics and the explanatory visualisations into a unified export. These features were excluded in favour of keeping the implementation lean and that it aligns with the project's current objectives.

5.7 Conclusion

This chapter has elaborated on the practical implementation of the system design from the prior chapters. Each module of the Shiny app was developed with the purpose of serving a specific function, starting from preliminary background operations such as launching the app, importing and loading the datasets, and constructing the monthly modelling panel. The following sections outlined how the Dashboard Overview and the Data Explorer tabs facilitate with exploratory analysis, how the Spatial Statistics tab connects geospatial diagnostics, and how the Predictive Insights tab integrates machine learning models with scenario controls for forecasting. Ultimately, the final module is the About & Download tab that provides transparency and reproducibility by offering in-app documentation and the ability to download datasets and outputs.
Throughout the system, the implementation is emphasized for the modularity, error handling, and extensibility. Data pipelines were crafted to verify input validation and provide clear and descriptive feedback when resources were missing, and the modelling framework compromised between interpretability with GLM and forecasting ability of XGBoost. Each tab was connected through a uniform workflow that moves from descriptive summaries, through diagnostic assessments, and projecting predictions.

In summary, the system's implementation illustrates how a technically complex system can be transformed into an intuitive decision-support system for insurers and policymakers. By integrating combination of geospatial analysis, machine learning, and interactive visualisation within a unified architecture, the system not only replicates the historical patterns but also supports the informed exploration of opportunities in the future. This established the foundation for both the testing and evaluating in the next chapter.

6. Testing and Evaluation

6.1 Testing Approach

Testing was performed at various levels to verify the accuracy of individual components as well as the overall reliability of the system. The process included combining unit testing of the helper programs, integration testing across different datasets and models, as well as end user testing of the interactive dashboard.

6.1.1 Unit Testing

Each helper functions were tested in independently to verify whether they produced the correct outputs. This include the spatial functions that calculate Moran's I that is responsible for

generating the neighbour lists, along with the data loaders that normalised the names of the counties and interpreted the dates. These checks and validations confirmed that the low-level functions were correct and reproducible, minimising the chance of underlying errors that could potentially affect the high-level modules.

6.1.2 Integration Testing

After validating the functions individually, integration testing assessed how they interacted. The monthly panel builder underwent testing with raw claim, climate, and disaster datasets to verify and ensure that the aggregation and joins produced consistent monthly records. Likewise, the outputs from the modelling pipeline were similarly tested to verify whether they matched the expected schema for subsequent visualisation and error analysis could be done later on. This phase also validated the helper were appropriately referred within the Shiny modules.

6.1.3 Dashboard Testing

The dashboard was evaluated from the perspective of the end users. The dashboard's tabs were tested on their responsiveness, whether it was displaying the correct plots and tables and their ability to handle input without becoming unresponsive. Some of the evaluations involved navigating across various tabs and selecting extreme parameters and transitioning between datasets and models. This assessment assisted in ensuring that the system can be operated effectively under realistic usage scenarios.

6.1.4 Error Handling Checks

The error handling was evaluated explicitly by deleting the files and testing the invalid parameters. The system provided clear and informative error messages instead of just failing without any indication. For instance, when there was inadequate data for calculating the Moran's I, the interface would display a notification rather than generating any misleading information that could confuse the user. This further instils confidence that the system would operate how it supposed to even when under pressure.

6.2 Evaluation of Predictive Models

The predictive features of the system were tested and evaluated in an effort to ascertain how well the selected models captured the insurance claims dynamics. The assessment emphasized that both the absolute performance as well as the interpretability performance were dealt with using the metrics that corresponds the data properties.

6.2.1 Metrics Chosen

There were four main metrics were used for the predictive modelling:

• **Mean Absolute Error (MAE):** this metric measures the average size of errors in their original units, so the results can be easily interpreted.
• **Weighted Absolute Percentage Error (WAPE):** This metric returns a scale-free ratio of error relative to the actual values, which is beneficial for evaluating the performance across different counties.
• **Bias Ratio:** This metric identifies the systematic over-prediction or under-prediction by

comparing the totals of total predicted amounts with totals of actual claims.
• **Zero-Hit Rate:** This metric evaluates the effectiveness of the models in predicting months with zero claims accurately, which is particularly important due to the data's heavy zero inflation.

Together, these metrics provided a combination of comprehensive understanding of the average accuracy, relative error, systematic bias, and detection of absence of events.

6.2.2 Limitations

Alternative metrics like Tweedie deviance or Poisson deviance were considered. These metrics are frequently favoured in the context of actuary as they directly and accurately represent the likelihood functions employed in model estimation. Nonetheless, they were omitted in this context since their meanings are less clear for non-expert users such as policymakers or insurers utilising the dashboard. In the same way, RMSE was used solely for amounts and not for counts because its significant penalisation on large spikes is less applicable in the count data which is dominated by zeros. The compromise involved placing greater importance on clarity and transparency for end-users rather than in depth actuarial statistical accuracy.

6.2.3 Error Analysis by County

The evaluation was extended beyond the general metrics to include spatial and hazard specific performance and effectiveness. The county level backtests were conducted to demonstrate that both the GLM and XGBoost were implemented to manage the quiet counties which are predominantly zero claims, which is clearly evident by the zero-hit rates. Nevertheless, the counties that had experience frequent wind and flood events (which is predominantly storm event when combined) the models differed, where the GLM was more accurate and effective at capturing the infrequent peaks due to its distributional assumptions, whereas the XGBoost was much more cautious at predicting therefore underestimating the significant events.

6.3 Results and Discussion

6.4.1 Model Performance

Model	MAE	WAPE	Bias	Zero-hit
GLM Poisson	0.080	2%	100%	0.976
XGB Poisson	0.164	2%	100%	0.978

Model	MAE	WAPE	Bias	RMSE
GLM Tweedie	4709.98	2%	100%	70055.28
XGB Tweedie	631.23	2%	100%	10761.46

This evaluation compared the GLM and XGBoost in terms of claim counts and amounts. For the counts, the GLM Poisson model had achieved a lower MAE score of about 0.08 compared to the

XGBoost Poisson model, which scored about 0.16. This indicated that the distributional assumptions of the GLM are more suited for a zero-inflated structure of the dataset. Both models were good at spotting months with no claims, getting close to 98% correct. When predicting the claim amounts, the results were different. The Tweedie GLM model recorded higher MAE values of approximately $ 4,700 but was able to reproduce spikes in the data. On the other hand, the Tweedie XGBoost had much smaller errors and an approximate value of $600 because of predicted the values closer to zero. The RMSE further emphasised this disparity, where the GLM model was heavily penalised for missing and failing to account for extreme spikes, incurring costs of $70,000, whereas the XGBoost had a greater stability with a penalty of only about $10,000, but the risks were underestimated for those rare but expensive events.

6.4.2 Visual Outputs

The dashboard visualisation verified these findings. Backtest graphs indicated that the GLM was more responsive to sudden increases although it did overestimate during silent months. XGBoost did produce smoother curves that were predominantly centred around lower values however it resulted in reducing the number of false positives, yet it had struggled to predict significant surges. The visualisations for the stress control effects, where the precipitation or windgust multipliers had added to the forecasted amounts in the future months. The table located in the forecast tab facilitates with a comparative analysis of the outputs from both the models which noted the perceived risk under each scenario. These visual representations ensured that the findings were to remain clear and effectively bridge and fill the gap from the technical modelling through a more practical advice support.

6.4.3 Spatial Evaluation

Spatial Statistics tab's residual analysis provided and offered another extra layer of diagnostics insights. Residuals are bunched up positively in certain counties, which suggests the presence of unaccounted regional influences such as local climate micro-patterns or reporting biases. Counties with frequent exposure to wind events consistently demonstrated systematic under-prediction by both GLM and XGBoost. The Moran’s I statistics further confirmed that residuals were not always strictly distributed, indicating that spatial features can be dealt with in future modelling more efforts. This spatial evidence served to enhance and understand the context to the error metrics that purely numeric summaries cannot provide and fail to convey.

6.4.4 Overall Assessment

The findings reveal a trade-off between traditionalism and interpretability. The general actuarial practice was more closely followed by making the use of GLMs, as it directly modelled the claim distribution and effectively captured rare events. However, penalties were higher when spikes were overlooked. XGBoost produced lower error rates by remaining close to the majority of low value months, yet it posed a risk of being underrepresenting the exposure to significant losses. From a decision-support standpoint, displaying both the models in concurrently with each other allows the user to weight the two viewpoints based on the risk tolerance. The combination of the error metrics, visual outputs, and spatial diagnostics provided a rounded evaluation of the system's performance, illustrating the merit of blending predictive analytics with interactive exploration.

7. Critical Review & Reflection

7.1 Achievements

The project achieved its goal by successfully developing and launching an interactive R Shiny dashboard that integrated climate, natural disaster, and simulated insurance claim data at the county level. The system integrated geospatial analysis with predictive modelling under one platform which sets it apart from typical industrial tools that keep visualisation and modelling separately. The system utilises both Poisson and Tweedie GLMs and XGBoost models to demonstrate how the prototype achieves balance between interpretability and predictive capabilities. The spatial statistics module further allowed users detect patterns of clustering and spatial biases while the predictive insights page provided scenario-based forecasts with stress controls in an adaptable manner.

From a technical standpoint, the modular architecture achieved success because the helper scripts operated independently to facilitate a clear source code that the developers could be reusable and maintainable to simplify debugging. The development of the monthly modelling panel represents as another milestone, as it integrated claims, climate and disaster data into a coherent analytical framework. Additionally, the prototype incorporated user friendly features which includes downloadable datasets, help text and view results that merged quantitative metrics like MAE, WAPE, RMSE, bias ratio and zero hit rate together with intuitive visualisations.

Ultimately, the primary accomplishment was the development of a prototype decision support system which allows insurers and policymakers to investigate the risk patterns and test and evaluate their assumptions under various climate and disaster scenarios.

7.2 Obstacles & Constraints

Numerous challenges had emerged throughout the development process. A significant hurdle was the data integration where standardising and normalising the names and time formats between disaster records from counties, climate and claim data. The disaster dataset contained unclear, missing and complex records like the ‘Admin Units’, which had resulted in an imperfect county attribution. Likewise, the climate data which needed to be aggregated and compute the lag proved to be rather difficult.

Another limitation was the modelling process. The dataset exhibited extreme zero inflation because a majority of the months recorded no claims. Although the GLM was able to effectively capture the distribution structure, whereas XGBoost showed a tendency to lean toward lower values, complicating the identification of the spikes. Furthermore, the process of hyperparameter tuning required considerable time and computational power which further pushed to be careful and balance the project timelines.

From the dashboard usability standpoint, the dashboard has been developed without direct participation of industry professionals. This limited approach can hinder the ability to gather genuine end user requirements, which led to having evaluation ne assessed based on heuristics and limited feedback instead of a comprehensive trail with insurers and policymakers. Ultimately, the timeframe for this project was very limited meaning many functionalities that were deemed important have been shelved and figuring out which functionalities to implement to create the most valuable product within limited timeframe was an accomplishment in itself.

7.3 Lessons Learned

During this project, it was emphasized that data preprocessing together with feature engineering form the fundamental building blocks for the completion of the system. A considerable amount of effort was dedicated into guaranteeing that the monthly panel was reliable, robust and consistent, highlighting the fact that the success of data science projects frequently depends on the preparation rather than the modelling techniques.

Another vital takeaway was the importance of a modular design. By implementing these divided individual functional components of the dashboard, the development had become much more manageable, helpful and simple. The debugging process had also become much easier as it became easier to figure out where the error happened. This approach helped understand the practices of professional software development, where the system prioritises maintainability and scalability.

Furthermore, the project demonstrated that statistical precision needs to be balanced with user friendly features that are easy to use. For instance, metrics like WAPE and bias ratio provide useful and valuable insights for technical end users, however they can perplex non-technical end users. Thus, there is a need to have explanatory notes and visual cues to assist when necessary. This experience demonstrated that technical precision must be complement by an effective communication method for decision support tools to achieve their intended impact.

Finally, the project established that a proper performance and realistic expectations for machine learning models when working with sparsed data. In situations where the dataset is zero heavy and have infrequent extreme event, no model exits that can predict spikes with certainty. This shifts the objective to offering a reasonable baseline and scenario-based explorations rather than striving for that flawless accuracy.

7.4 Improvements & Future Work

The current prototype show that the geospatial analysis and predictive modelling can be integrated together to act as decision support tools for the insurance industry, however it could still use various enhancements that could improve and boost the systems reliability, robustness, and practical utility. One significant area exists in the form of data quality and coverage. The system relies on the refined yet static CSV datasets that uses simulated insurance claims to cover the gaps. Although this method allowed to meet the project deadline, but the future research needs to include real claims data that needs to be anonymised and protected and managed under suitable data sharing agreements along with a validated historical disaster loses dataset. This integration would strengthen the foundation of forecasting and better align the outputs with the requirements of the industry.

There is potential for further diversification of the predictive models. Although Poisson and Tweedie GLM and XGBoost offer advantages, future developments branching into the application of Long Short-Term Memory (LSTM) networks for learning temporal sequences, or the Bayesian hierarchical models could account for the uncertainty at the county level. Additionally, spatial statistics could further be explored beyond the Moran’s I and into Local Indicators of Spatial Association (LISA) and Getis-Ord Gi* analyses, which would facilitate hotspot detections on a more granular scale. These tools will allow the insurer and policy makers to identify precise high-risk locations rather than observing general trends and patterns.

In addition to the modelling, the system architecture can further be improved. Currently the app exists only on the local systems which restricts both accessibility and scalability. As the system needs to transition and grow, implementing it on cloud-based environments like AWS, Azure or RStudio Connect would allow for a multi user access with role-based permission and allow for a scale on demand larger datasets. The deployment can become more efficient through the use of Docker or Kubernetes to help with maintenance and version control. Implementing this measure would advance this prototype towards a production ready and support which can be used by the insurance industry readily.

Data pipelines that utilise APIs can be established to facilitate future advancements. The system is capable of acquiring climate drivers via automatic updates sourced from the UK Met Office API, as well as receiving flood alerts from the Environment Agency and disaster records from EM-DAT. The monthly panel would receive automatic near real-dime updates from the real-time claim's feeds provided by insurers. The system ensures the maintenance of real-time forecasts through automated processes that perform data refreshes without the need for human involvement.

The proposed enhancements can lead to direct progress by improving data representation, research methodology, updating deployment strategies and improving user interface experience. The system would transform from a prototype into an operational tool which supports insurers and regulators and policymakers for real-world risk management through practical scenarios.

8. Conclusion

8.1 Summary of Work

This project aimed to design and developed an interactive R Shiny dashboard for analysing and forecasting the financial effects of natural disasters on insurance claims in the UK. Multiple datasets were implemented, which comprised of UK and UK county climate, natural disasters and simulated claims. These data were transformed to establish the monthly panel to facilitate for both the spatial and predictive modelling operations. The application was organized into five modules, which included Dashboard Overview, Data Explorer, Spatial Statistics, Predictive Insights, and About & Download, ensuring a structured and user-friendly interface.

The predictive modelling utilised GLM and XGBoost (with Poisson and Tweedie) to generate predictions for claim counts and amounts. The evaluation process employed MAE, WAPE, RMSE, bias ratio and zero-hit rate metrics that indicated the data distribution characteristics of skewness and zero-heavy values. The interface received direct integration of Moran's I-based residual spatial diagnostics. The system allowed users to examine various "what-if" scenarios of weather-related factors through the scenario-based stress control system. The combination of these components resulted in a prototype system which merged statistical precision with interactive visualization tools to help insurers and policymakers evaluate and prepare for financial risks associated with climate change.

8.2 Final Reflection

The overall project demonstrated that a relatively well-designed prototype at a small scale can unite different data sources with modelling techniques and geospatial analysis to create a unified system for insurance risk evaluation. The dashboard serves as a proof of concept to demonstrate

how interactive data-driven tools assist insurers and policy maker in assessing climate-related financial risks, but it does not function as a product ready operational system.

This report shows that usability functions is an essential element for technical systems like this. Moreover, specialized and prior knowledge is required to understand predictive models independently, but they are more interpretable, and the system becomes more accessible through the use of visualisations and spatial diagnostics. The combination of statistical accuracy with user-friendly design follows the current data science direction which supports tools that explain data towards transparency, explainability, and user-centred tools.

The research establishes the limits of the system. The simulated claims dataset, while carefully designed, inevitably lacks the richness and complexity of real exclusive data. The predictive models used GLM and XGBoost families, but alternative architectures including recurrent neural networks and Bayesian hierarchical models would offer more insights when working with bigger datasets. The usability assessments employed a heuristic method instead of involving various insurance professionals meaning it would provide with better understanding of what the system actually lacks and can improve on.

The project successfully achieved its goals by integrating various data sources and employing spatial and predictive methodologies to develop an operational interactive system that serves both academic research and disaster risk management within the insurance sector. Future investigations will build upon this foundation by analysing extensive datasets, exploring diverse modelling techniques, and conducting real-world field tests in collaboration with insurance companies.

In summary, the project offers a systematic, evidence-driven illustration of the integration of machine learning, spatial analysis, and interactive dashboards to enhance decision-making in a sector that is progressively influenced by climate risk. Its contributions are significant not only in the technical outputs that were generated but also in demonstrating how applied data science can connect academic research with practical industry challenges.

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10.Appendix 1: Project Proposal

1. Overview

Natural disasters have a profound effect on the insurance sector, resulting in an increased surge of claims, heightened premiums rates and financial pressure on both insurers and policyholders. The (Swiss Re Institute, n.d.) reported that in 2023, natural catastrophe resulted in global economic loss amounting to $280 billion, of which more than 50% were not covered by insurance. Recent data statistics from The Association of British Insurers reveals that the total claims payout for the year 2024 was £4.1 billion setting a record-breaking amount to be paid out in the first nine months of any year (ABI, n.d.). It was also reported that in the third quarter of 2024, claims regarding damage to residences resulting from storms, significant rainfall, and frozen pipes amounted to £136 million, whereby marking the sixth consecutive quarter where weather related claims exceeded £100 million. Furthermore, this is a 6% increase compared to the £128 million disbursed in the same quarter of 2023. The (GOV.UK, n.d.) reported that due to the impact of climate change, it is projected that there will be an increase in the total number of properties located in regions that will be vulnerable to flooding from rivers, sea, or surface water, potentially reaching 8 million properties (increased from 6.3 million properties) by mid-century. Implying that 1 in 4 properties will be at risk of such flooding.

This project proposes to create an interactive dashboard designed to analyze and forecast the effects of natural disasters and catastrophes on insurance claims within the United Kingdom. By leveraging historical data on disasters, information regarding insurance claims and geospatial mapping techniques, the application attempts to deliver practical insights concerning claim trends, regional risk assessments and financial consequences. The development of this dashboard represents a significant contribution to the ongoing efforts aimed at reducing the financial risks connected to disasters faced in the UK, thereby facilitating improved preparation, prevention, and response approaches within the insurance industry. This emphasizes the critical necessity for tools that can effectively comprehend and manage such events. The project is a student-initiated project that will try to tackle the critical issues that insurers encounter in the efficient management of claims arising from disasters.

This research will require gathering and preparing datasets from credible sources, including the UK Met Office, The Environment Agency and simulated insurance claims. Various analytical methodologies, such as geospatial analysis and machine learning will be used to discern patterns and develop predictive models. For instance, using clustering algorithms and time-series forecasting techniques will help to estimate future claim volumes in relation to disaster severity, geographical location, and historical trends. Studies by (Manfré et al., 2012b) highlights the idea of having geospatial database infrastructure to generate information which will be useful for preparedness and risk management strategies. Furthermore, machine learning techniques presents a transformative opportunity within the insurance industry for its ability to leverage powerful algorithms to analyze vast amounts of data which can help the insurers to gain a deeper insight and deliver a more efficient customer-centric insurance experience explains (Kasaraneni, 2020b) by harnessing these approaches, the project aims to create a dashboard that combines machine learning analytics together with interactive visualizations by disaster types, geographical region and time to obtain a deeper understanding into potential financial implications.

This project is interdisciplinary by integrating elements from data science, geospatial analytics, and financial modeling. It adds to the MSc Data Science (Fintech) program by using sophisticated machine learning and visualization techniques to tackle a real-world problem. Modules like Machine Learning and Applied Machine Learning will facilitate the formulation of machine learning models whereas Data Visualization and Big Data will assist in developing a user-friendly interface. Additionally, the project satisfies the QFC guidelines by merging theorical research together with practical applications, thereby promoting both academic integrity and relevance to the industry.

2. Objectives (181 Days)

2.1 Examining historical data related to natural disasters and their association to insurance claims. (36 Days)

- Activities
    - Undertaking a thorough literature review that explores the relationship between natural disaster and insurance claims by observing patterns, existing challenges, and emerging trends. (20 Days)
    - Analyzing the collected datasets to natural disasters such as floods and storms and corresponding insurance claims from reputable sources. (10 Days)
    - Identifying gaps in existing literature and identifying opportunities where predictive modeling and data visualization could enhance understanding and decision-making. (6 Days)
- Deliverables
    - The literature review in the final report will encapsulate existing research, industry practices and the necessary enhanced tools in disaster insurance analytics.
    - Developing a clean and organized dataset including disaster insurance claims and geospatial data.

2.2 Analyzing the patterns and relationship within the consolidated dataset.

- Activities
    - Conducting exploratory data analysis (EDA) aimed to uncover the trends in claims such as surges in claims after a disaster or seasonal changes. (20 Days)
    - Performing geospatial analysis to identify regions that are most impacted by certain disasters and their claim volume. (10 Days)
    - Extracting features that are relevant to the machine learning model. (5 Days)
- Deliverables
    - All the data visualization like heatmaps and charts will be presented in the final report.
    - A thorough analysis chapter that explains the results and insights, as well as their implications for system design.

2.3 Create and execute a machine learning model for estimating both the volume and cost of claims.

- Activities
    - Create machine learning models such as regression analysis, random forest algorithms or ARIMA models to predict claims. (15 Days)

    - Training the models with historical datasets and testing for their accuracy with metrics like RMSE and precision. (10 Days)
    - Documenting the model, the selection process and the explanation for the strategies used. (6 Days)
- Deliverables
    - Get a fully trained machine learning model that can accurately predict claims under various disaster situations.
    - The final report will describe the model development process, assessment measures and outcomes.

2.4 Creating an interactive dashboard aimed to analyze the impacts of disaster on insurance.

- Activities
    - Making a user-friendly interface that includes filters for disaster types, geographical region, and time periods. (15 Days)
    - Adding all interactive visualizations and machine learning models to the dashboard. (10 Days)
    - Implementing functionality to allow users to enter specific disaster scenarios to produce relevant predictions. (6 Days)
- Deliverables
    - Completed dashboard with all visualization capabilities and predictive analytics.
    - A section in the final report that describes the design and execution of the dashboard with screenshots of the architectural diagrams of the system.

2.5 Testing and evaluating the system.

- Activities
    - Testing the dashboard and machine learning model to verify accuracy, usability, and performance metrics. (10 Days)
    - Obtain input from the supervisor on the system operational capabilities and overall effectiveness. (8 Days)
    - Refine the system in response to the feedback to solve any flaws and increase usability. (6 Days)
- Deliverables
    - Include a testing section in the final report that details the test cases, outcomes and any modifications made to the system.

2.6 Compile all the findings and prepare the final report.

- Activities
    - Prepare the final report including sections on literature review, system design and implementation, testing procedures and evaluation metrics. (10 Days)
    - Present a professionally written conclusion and future recommendations for prospective research. (8 Days)
    - Ensure report meets all academic standards and contains all relevant references to used sources. (6 Days)
- Deliverables
    - Complete the final report with all the sections.
    -

3. Legal, Social and Ethical Issues

3.1 Legal

The project will use information from natural disasters and insurance claims, which necessitates careful consideration of the UK Data Protection Act of 2018 (DPA) and the General Data Protection Regulation (GDPR). These legal frameworks demand that any personal or sensitive data must either be anonymized or pseudonymized to safeguard the identities and privacy of individuals. To maintain compliance, the project will use an anonymized dataset or simulated data that omits any personally identifiable information.

The utilization of datasets that may be obtained from public or licensed platforms will require strict meticulous consideration and adherence to copyright and licensing regulations. For instance, any dataset that is available under the Open Government License (OGL), which allows free usage if proper attribution is provided. Adhering to these attribute requirements and upholding all intellectual property rights is essential in ensuring the project's legal integrity.

Considering the interactive characteristics of the dashboard and the possibility of handling sensitive information, the project will comply with recognized security protocols. As such, when the dashboard is deployed online, mechanisms to combat unauthorized access, injection flaws and insecure data access controls will be required to be established to avert unauthorized use. By following these standards, the system will satisfy most of the legal obligations, but also foster confidence in its design and execution.

3.2 Social

The project will ensure that it meets the societal norms, notably in terms of accessibility and risk of misunderstanding and misinterpreting any potential outcomes. Accessibility is a fundamental component in this project, since it will be used by a wide range of users which includes individuals with disabilities. The project will comply with Web Content Accessibility Guidelines (WCAG), which provides a framework for guaranteeing digital accessibility. This involves incorporating features such as high-contrast settings, reader compatibility and intuitive navigation to support users with visual or motor challenges. By prioritizing accessibility, the project not only adheres to the Disability Discrimination Act of 1995 but also embraces the wider notion of inclusiveness in technology.

If users lack a comprehensive understanding of the tool's limitations, the machine learning model and visualization may cause misunderstanding. For instance, predictors are fundamentally probabilistic and should not serve as the key choice when determining insurance premiums or distributions of resources. To address this concern, the system will incorporate explicit warnings about the machine learning model assumptions, data constraints and biases. This will keep users well informed and help minimize unexpected outcomes, such as excessive dependence on predictions in the decision-making process.

Another social issue is the capability of the tool to expose regional disparities in disaster planning and response. By pinpointing regions that experience greater claim volumes or expenses, the system can illuminate areas that are neglected thereby encouraging more equal support allocation. Nevertheless, it is important to exercise caution to prevent labelling certain regions as high risks in a manner that might result in unjust treatment, such as higher insurance premiums for inhabitants of those areas. The application needs to prioritize to communicate the finding of the tool to promote fairness and equity.

3.3 Ethical

Predictive models are fundamentally based on historical data, which can often harbour intrinsic biases. For instance, insurance claims related to disaster may be skewed towards areas with better insurance coverage, thereby neglecting regions that are underserved or underinsured. Failure in recognizing and addressing these biases could result in predictions that perpetuate disparities or give incorrect information. To counteract this issue, the project will conduct a comprehensive study of the dataset to identify any potential biases and employ tactics such as balanced sampling and feature normalization to promote fairness in model result. Additionally, regular review of the model's predictions will aid in detecting and correcting any unwanted biases.

Transparency and clarity play a vital role in fostering user trust and encouraging ethical systems use. The machine learning models inherently operate on probabilistic principles and are based on assumptions that the users must understand in order to interpret the outcomes responsibly. Furthermore, the dashboard will feature comprehensive documentation and warning that describe how the process od prediction are made, constraints of the data and the extent of the insights presented.

The system will emphasize non-discriminatory methods by refraining from utilizing demographic or socio-economic factors that might result in biased treatment of certain groups and areas. For instance, forecast would exclude data elements that might contribute to discriminatory pricing for vulnerable people. Furthermore, the tool will be structured to promote fair resource distribution and disaster preparation as well as focusing on supporting marginalized regions instead of labelling them as high-risk.

4. Resources

To effectively execute this project, a variety of hardware, software and data resources are essential. The use of a dependable laptop equipped with 16GB RAM and multi-processing capabilities that can compute and handle data processing, machine learning and development of dashboard by making use of R and RStudio. Data processing, visualization and machine learning analysis will be carried out using the key R packages such as shiny, tidyverse, ggplot2, leaflet and forecast. Additionally, access to cloud storage or an external hard drive will enable secure backups, thereby mitigating the chance of data loss. The Literature Review and System Design will be researched through academic journals, articles and industry reports from the university library, IEEE Xplore or Google Scholar.

The project will require disaster and geographic datasets which can be obtained from the UK Met Office, the Environment Agency and OpenStreetMap. If genuine datasets are not accessible due to privacy concerns or licensing limitations, simulated insurance claims data will be generated. Furthermore, having a constant review of the progress from the supervisor will help resolve any difficulties quickly and keep project on schedule.

5. Critical success factors

The success of this project hinges on several essential aspects, including access to accurate datasets, appropriate utilization of software tools and referring to the supervisor. A possible risk pertains to the availability and quality of real-world data, which can be mitigated by creating

synthetic datasets based on industry patterns. Another problem is the technical complexity of integrating the R shiny dashboard together with machine learning models and interactive visualization. These technical concerns can be addressed by regular testing and debugging as well as using online platforms like R Studio Community or Stack Overflow. Time management is particularly important since delays in any step might jeopardize the projects timeframe. This can be solved by a comprehensive project plan which includes milestones and routine progress assessment with the supervisor.

6. Schedule

The start of the project will be on 13/01/2025 and the end date is 30/06/2025. There are micro deadlines for all the objectives which will be strictly followed (Theses are showed on the objectives). The first main deadline which is the literature review which is on the 10th of March 2025 and the interim report is on the 10th of May 2025.

7. Your references

ABI. (n.d.). *Year-to-date property claims payouts hit £4.1 billion | ABI*. Retrieved January 10, 2025, from https://www.abi.org.uk/news/news-articles/2024/112/year-to-date-property-claims-payouts-hit-4.1-billion/

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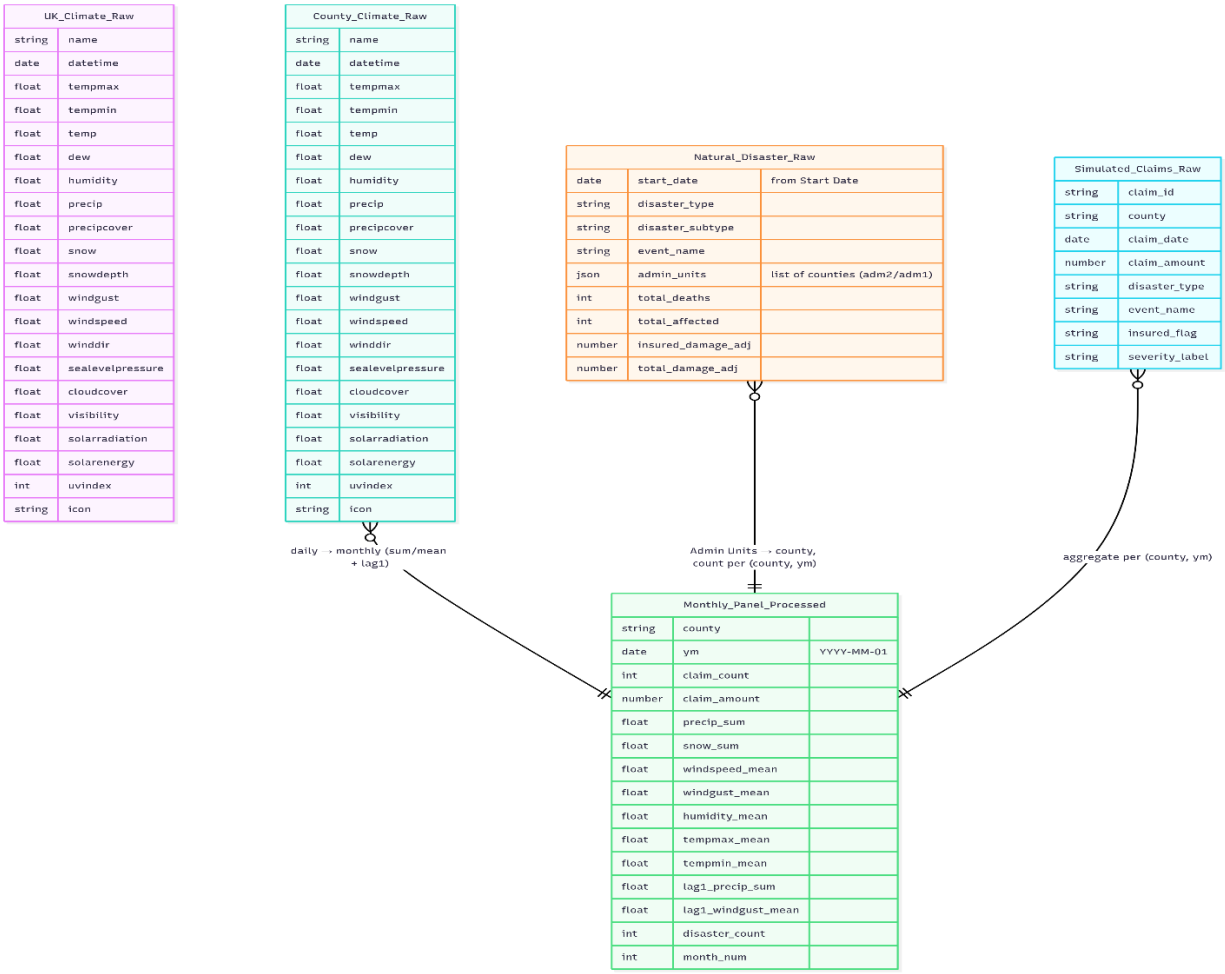
Manfré, L. A., Hirata, E., Silva, J. B., Shinohara, E. J., Giannotti, M. A., Larocca, A. P. C., & Quintanilha, J. A. (2012). An Analysis of geospatial technologies for risk and natural disaster management. In *ISPRS International Journal of Geo-Information* (Vol. 1, Issue 2, pp. 166–185). MDPI AG. https://doi.org/10.3390/ijgi1020166

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11.Appendix 2 Design Development Extra

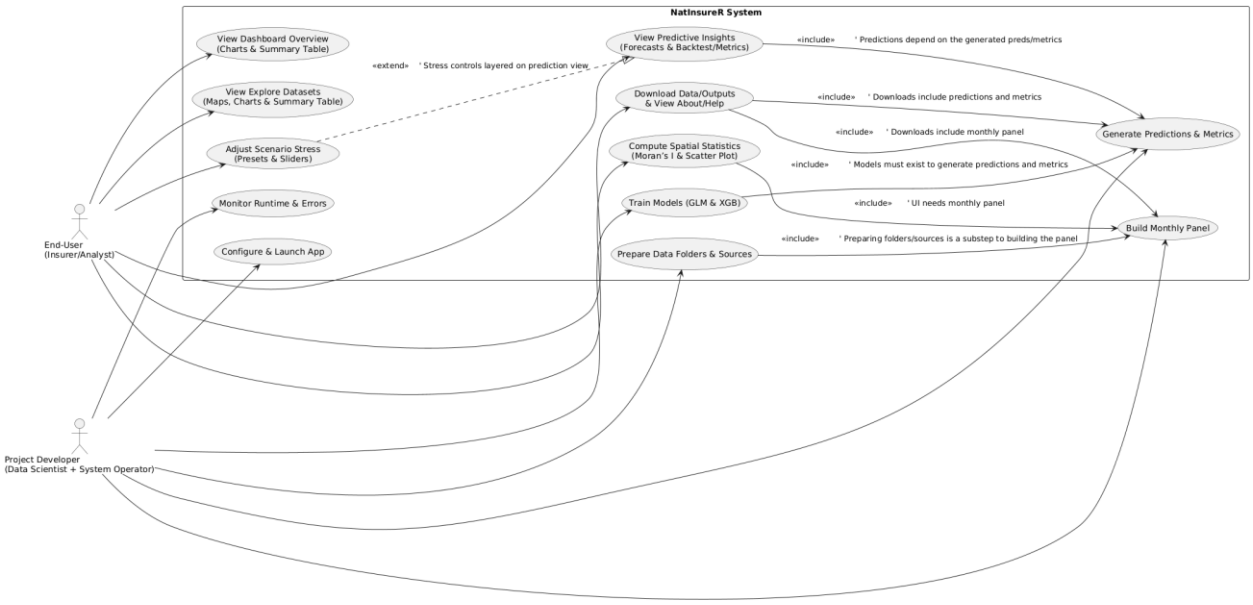
Entity Relationship Diagram

This entity relationship diagram depicts the process by which raw datasets concerning UK climate, county climate, natural disasters, and simulated insurance claims are preprocessed and integrated into a cohesive Monthly Panel. This panel connects county, time (ym), and derived features, therefore establishing the basis for predictive modeling and spatial analysis within the system.



Use Case Diagram

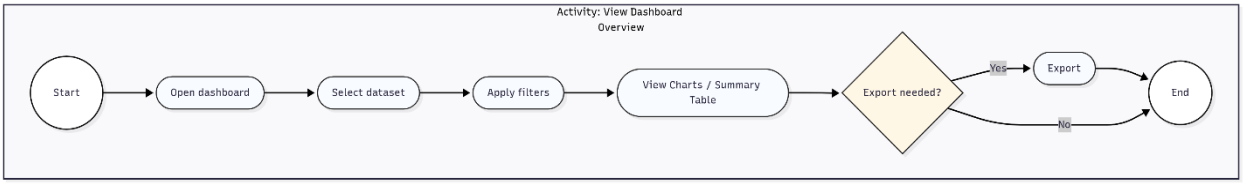
This use case diagram depicts the interactions between end-users and developers with the NatInsureR system, focusing on data preparation, panel construction, model training, and the access to dashboards, spatial statistics, and predictive insights.



Activity Diagrams

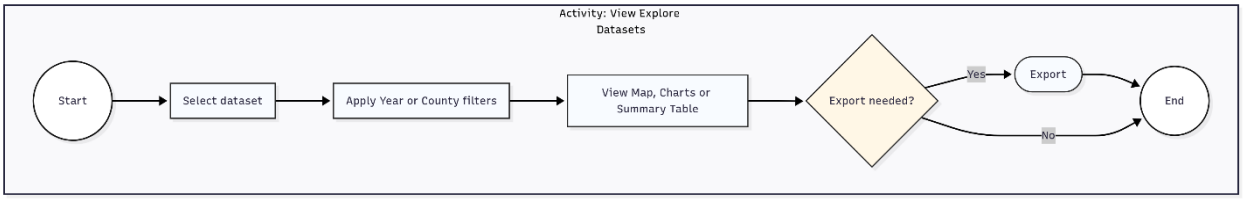
Dashboards Overview

This activity diagram illustrates the workflow of the end-user for accessing the dashboard overview. It begins with the user opening the dashboard and selecting a dataset, followed by the application of filters, the examination of charts and tables, and ultimately, the decision on whether to export the results.



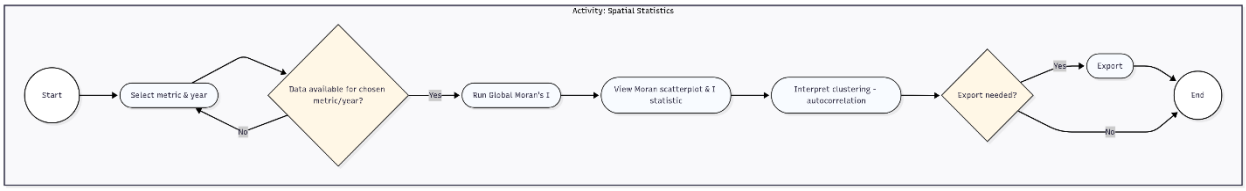
Data Explorer

This activity diagram depicts the manner in which users engage with the dataset explorer. Users can select a dataset, apply filters based on year or county, view various representations such as maps, charts, or summary tables, and have the option to export the results prior to concluding the process.



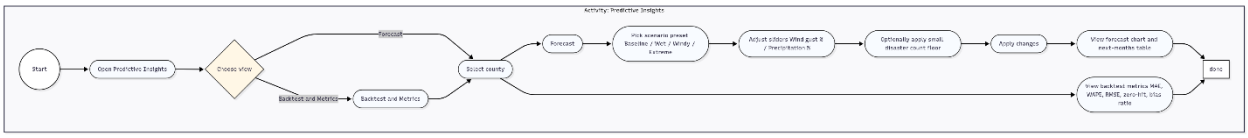
Spatial Statistics

This activity diagram illustrates the manner in which users engage with the spatial statistics module by choosing a specific metric and year, executing Global Moran's I when data is accessible, analysing clustering via the Moran scatterplot, and optionally exporting the findings.



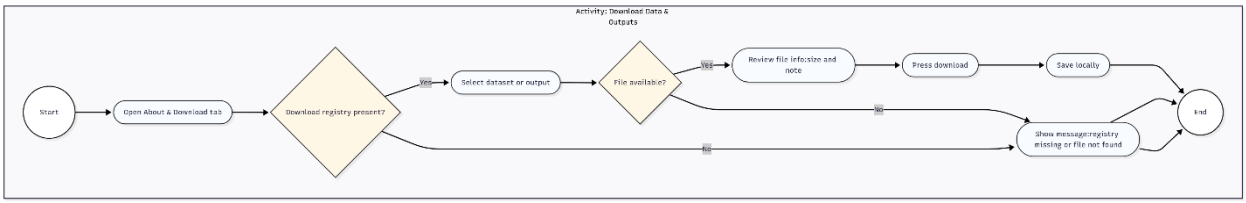
Predictive Insights

This activity diagram depicts the workflow for predictive insights, allowing users to opt for either forecast or backtest views. Users can select a specific county, modify scenario stress controls like wind and precipitation, optionally implement a disaster floor, and subsequently examine either forecast charts and tables or backtest metrics, which include MAE, WAPE, RMSE, zero-hit rate, and bias ratio.



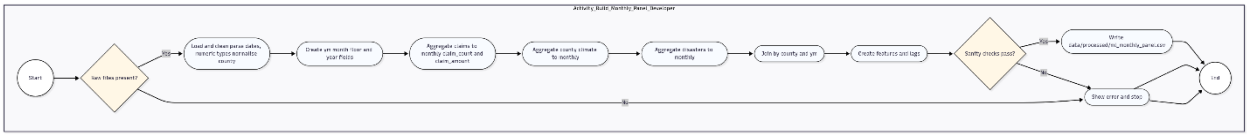
Download & About

This activity diagram illustrates the procedure for downloading datasets or outputs. The user initiates the process by accessing the About & Download tab, verifies the presence of the download registry, selects an available dataset, examines the file details, and subsequently either downloads and saves the file locally or encounters an error message if the registry or file is absent.



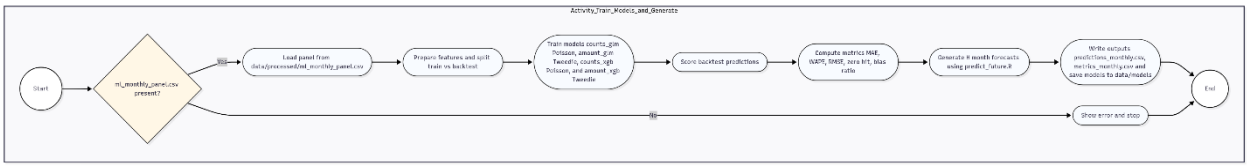
Building Monthly Panel

This activity diagram depicts the procedure for constructing the monthly panel, wherein raw files undergo validation, cleaning, and aggregation into monthly claims, climate, and disaster features. These features are then combined by county and month, followed by sanity checks prior to being saved as a processed dataset, or halted with an error if validation does not succeed.

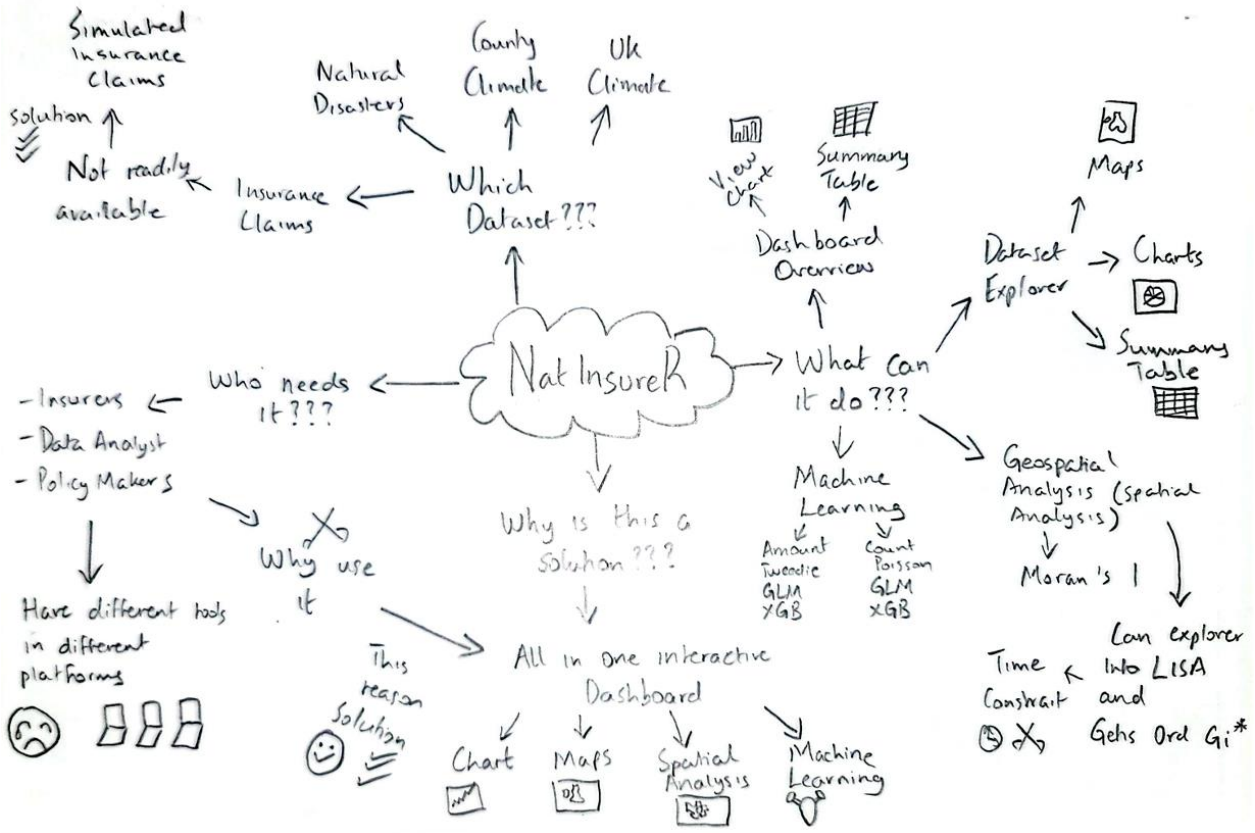


Training Models

This activity diagram depicts the workflow for training and generating outputs, wherein the system verifies the existence of the monthly panel, prepares the necessary features, trains the GLM and XGB models, assesses backtest metrics, produces forecasts, and either saves the outputs and models or halts with an error if any prerequisites are absent.



Rich Picture

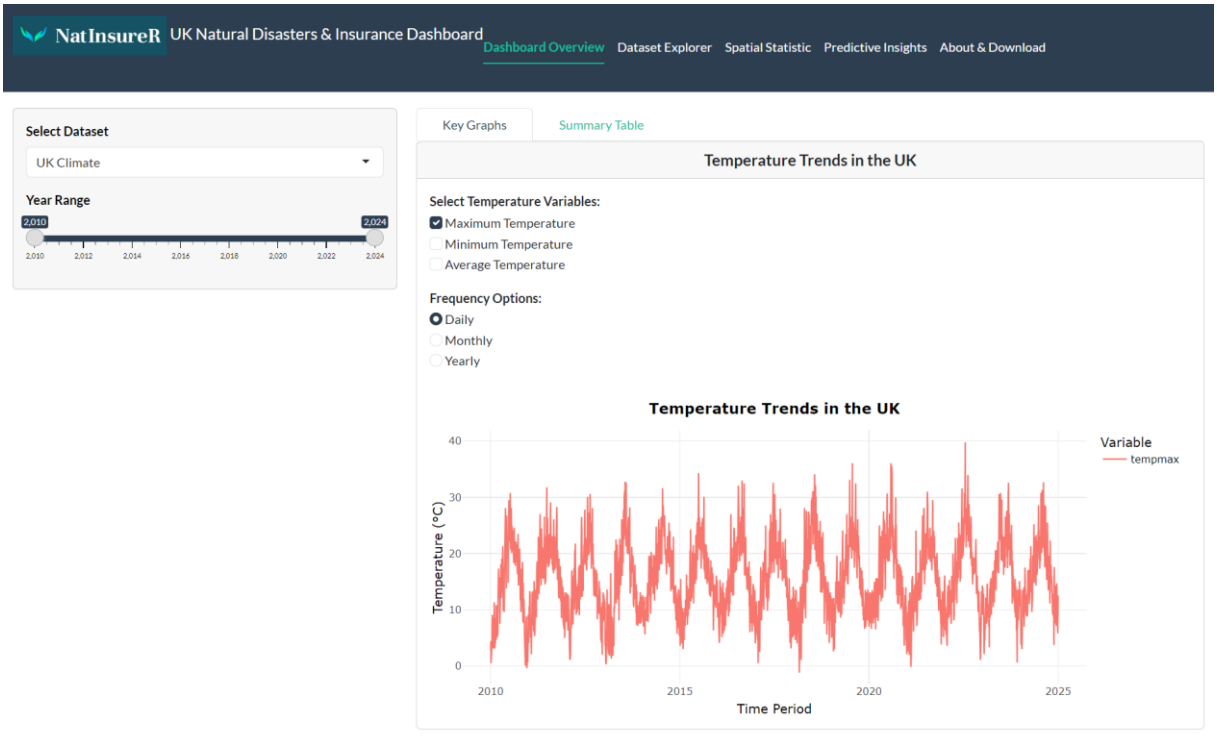


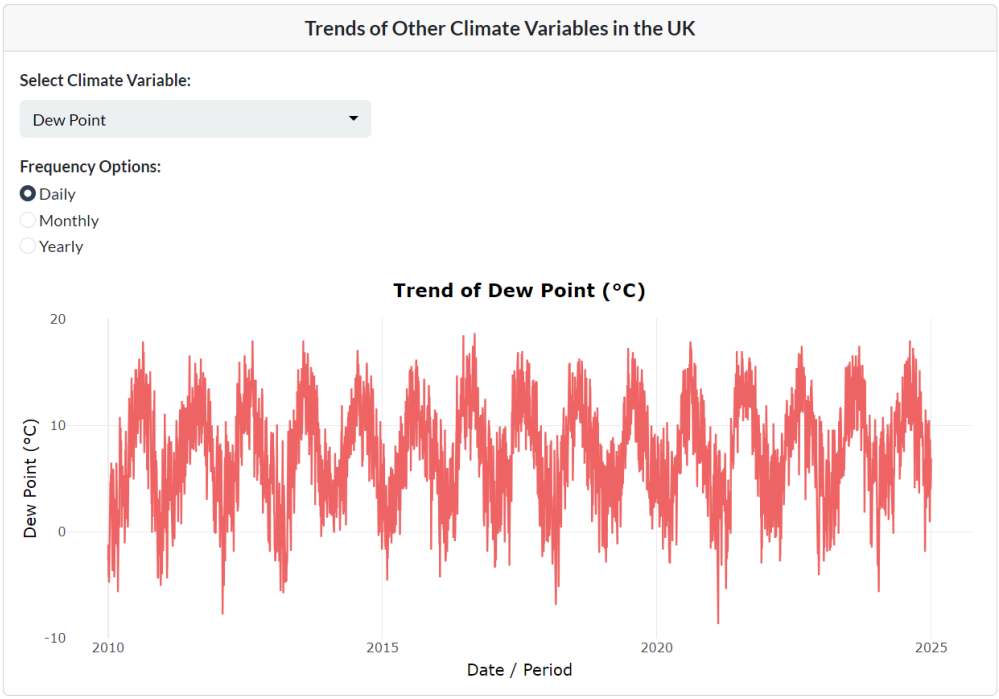
12.Appendix 3 Implementation

Dashboard Overview

In the Dashboard Overview, the user can choose 3 datasets (UK climate, Insurance Claims and Natural Disasters) and the year range. There is an overview of key graphs and summary tables for each dataset. The images below will depict all the graphs and summary tables for each dataset. (I will not show each plot with each variable, which would amount to many plots).

UK Climate Dataset

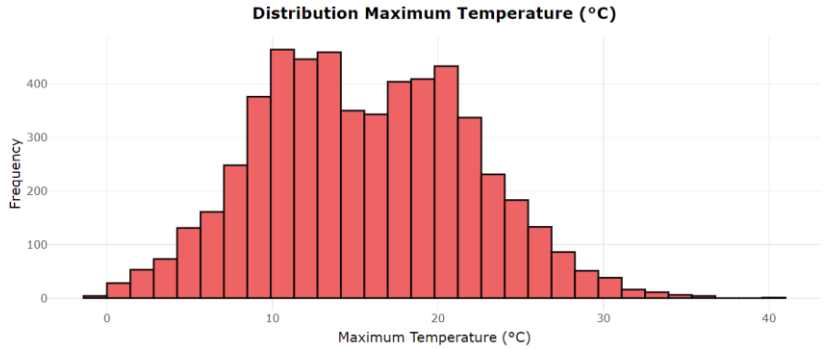




Distribution of Climate Variables in the UK

Select Climate Variable:

Max Temp



Precipitation and Snow Over Time in the UK

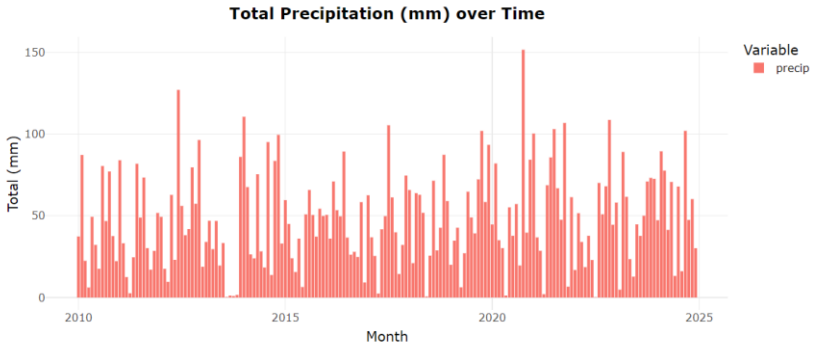
Precipitation and Snow Over Time in the UK

Select Variables:

- (•) Precipitation
- ( ) Snow

Frequency Options:

- ( ) Monthly
- ( ) Yearly



Risk Scatterplots

Select Scatter Plot:

Fog/Freezing Risk: Temperature vs Dew Point

Colour points by:

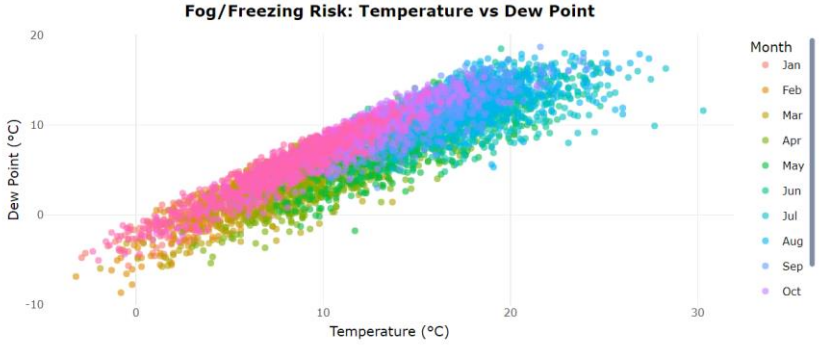
- (•) Month
- ( ) Year

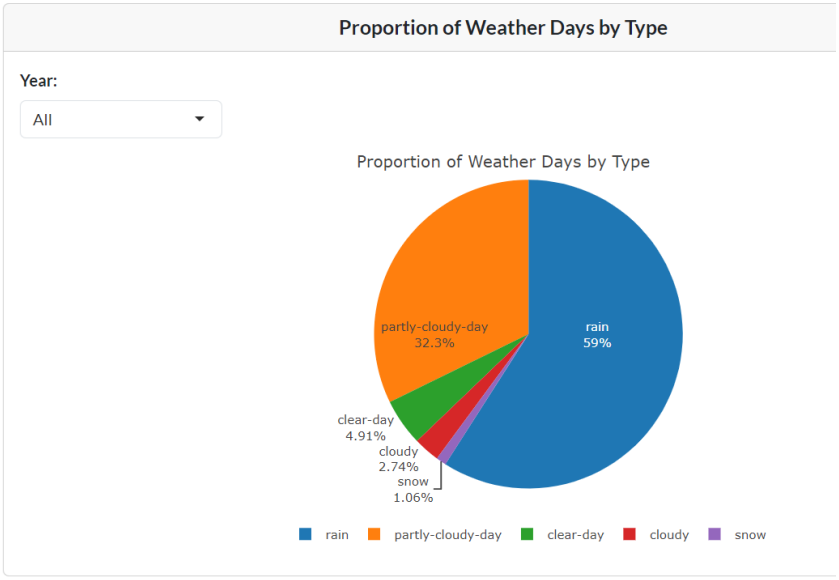
Filter by Month:

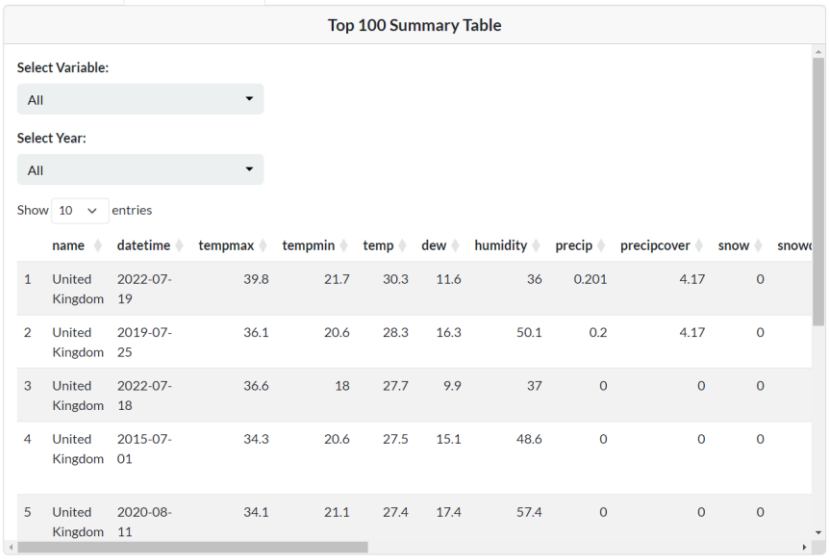
All

Filter by Year:

All

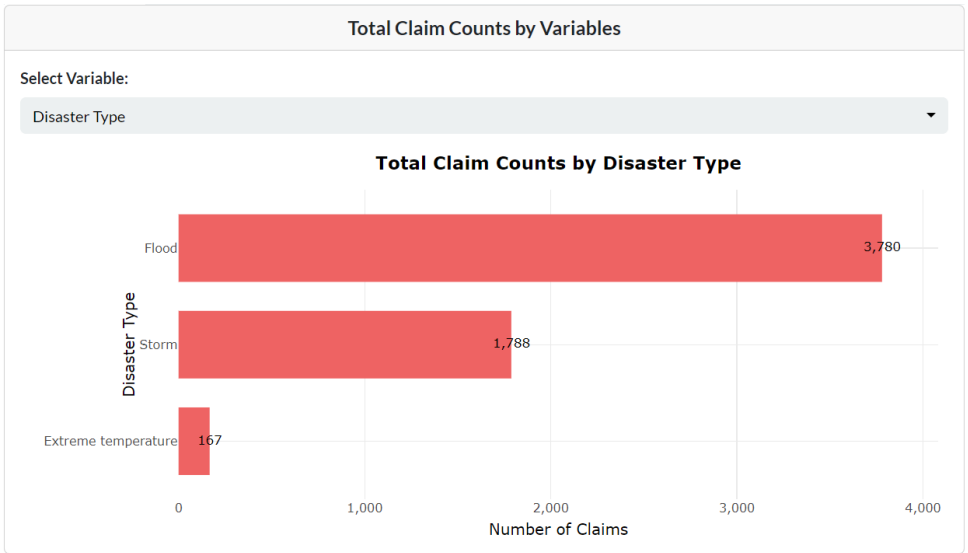






	name	datetime	tempmax	tempmin	temp	dew	humidity	precip	precipcover	snow	snow
1	United Kingdom	2022-07-19	39.8	21.7	30.3	11.6	36	0.201	4.17	0	
2	United Kingdom	2019-07-25	36.1	20.6	28.3	16.3	50.1	0.2	4.17	0	
3	United Kingdom	2022-07-18	36.6	18	27.7	9.9	37	0	0	0	
4	United Kingdom	2015-07-01	34.3	20.6	27.5	15.1	48.6	0	0	0	
5	United Kingdom	2020-08-11	34.1	21.1	27.4	17.4	57.4	0	0	0	

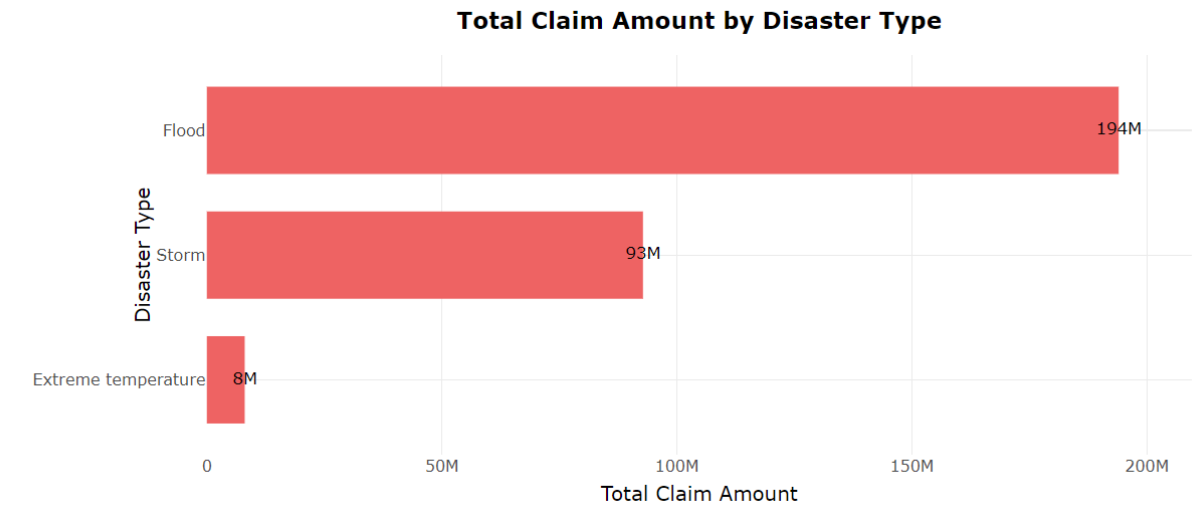
Insurance Claims Dataset



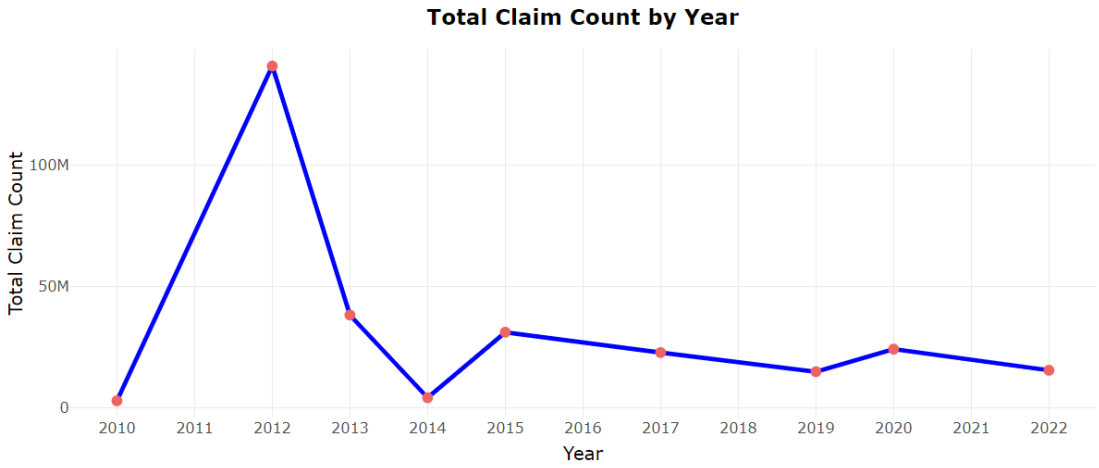
Total Claim Amounts by Variables

Select Variable:

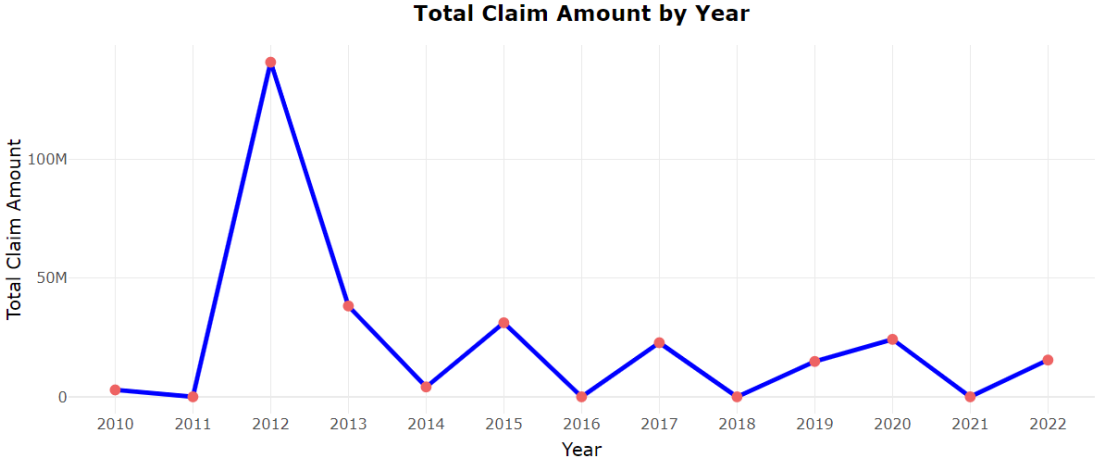
Disaster Type



Total Claim Counts by Year



Total Claim Amount by Year



Grouped Summary Table for Insurance Claims

Group by:

County

Show 10 entries

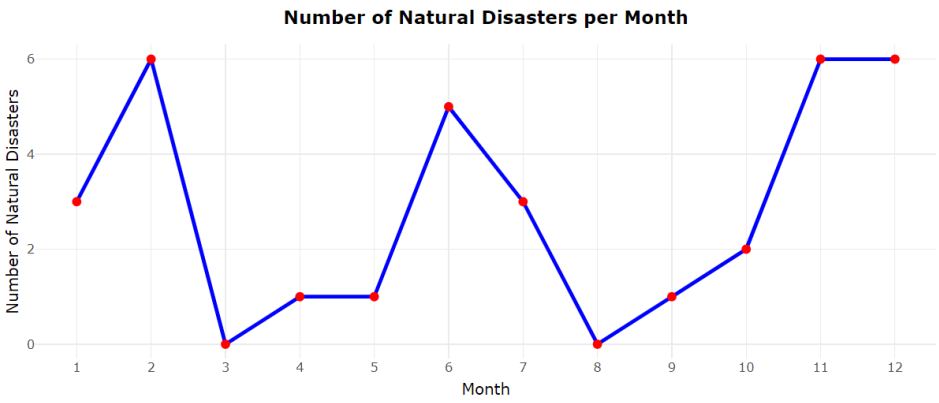
group	Total Claim Amount
North Yorkshire	30846700.6
Cumbria	27686001.21
Lancashire	25827011.6
Northumberland	21578861.74
Cornwall	13710703.77
Kent	12943043.89
Wiltshire	10850910.1
Isle of Wight	10810595.79

Natural Disasters Dataset

Disaster Frequency Over Time

Frequency Options:

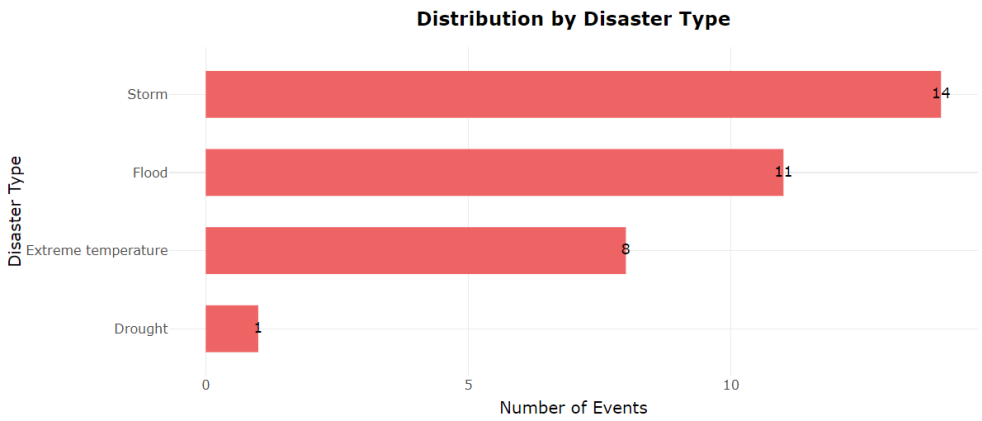
(•) Month ( ) Year



Distribution of Disasters

Select Plot:

(•) Disaster Type ( ) Disaster Subtype



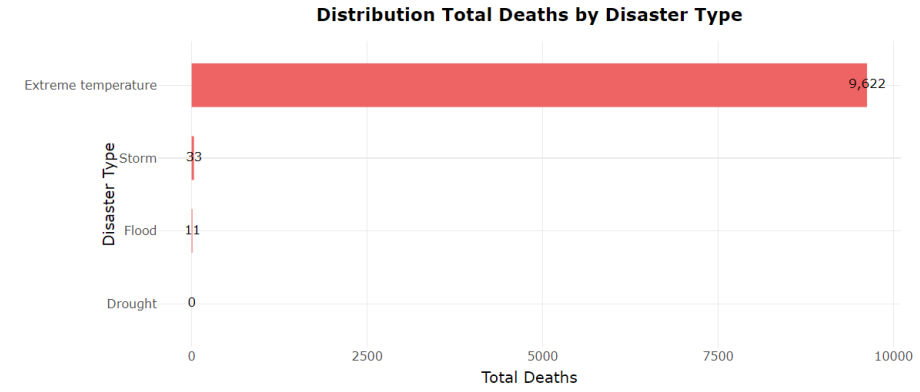
Distribution of Variables by Disaster Classes

Select Variable:

Total Deaths

Select Class:

- (•) Disaster Type
- ( ) Disaster Subtype
- ( ) Event Name



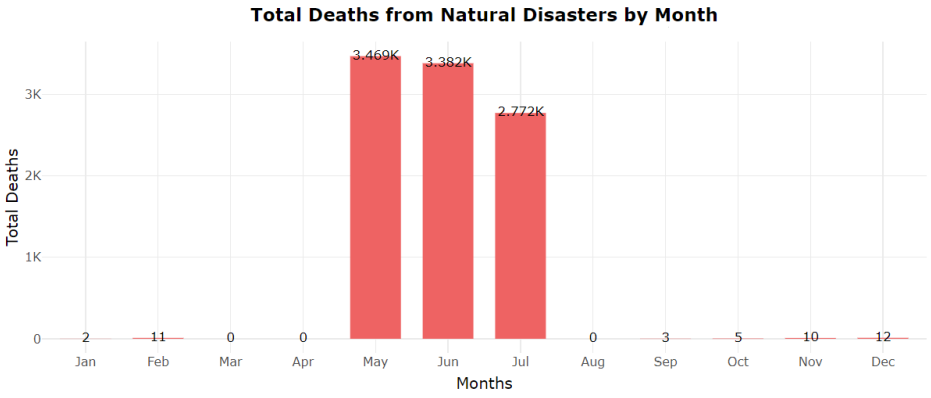
Distribution of Variables by Period of Time

Select Variable:

Total Deaths

Frequency Options:

(•) Month ( ) Year



Grouped Summary Table for Natural Disasters

Group by:

Disaster Type

Select Variable:

Total Affected

Show 10 entries

Disaster Type	Total Affected
Flood	53575
Storm	38554
Drought	0
Extreme temperature	0

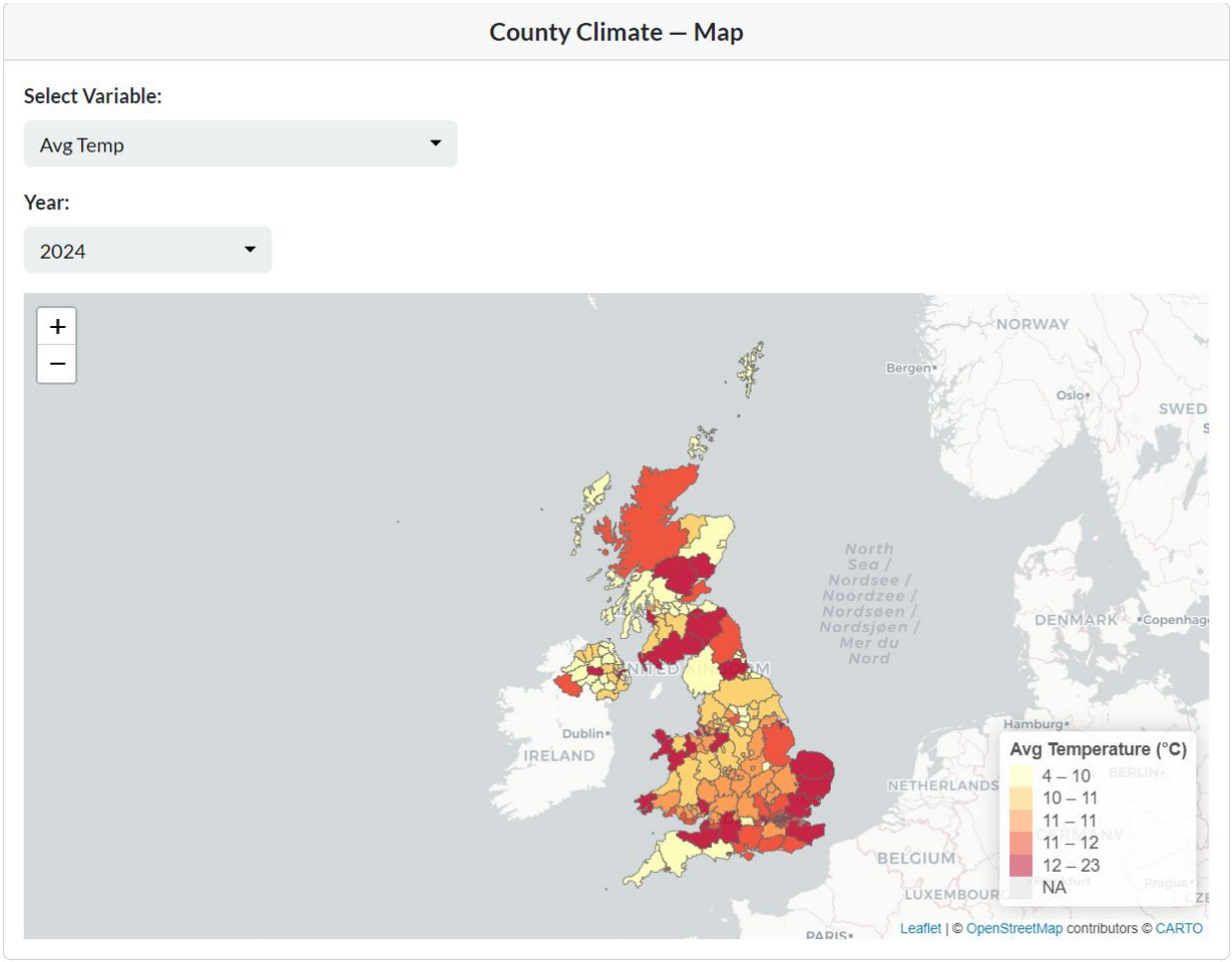
Showing 1 to 4 of 4 entries

Previous 1 Next

Dataset Explorer

This has the same idea as the Dashboard Overview, but has in dept graphs and summaries, plus it has an additional leaflet choropleth map. This uses 3 datasets which are County Climate, Insurance Claims and Natural Disasters datasets.

County Climate Dataset



Temperature Trends in the UK Counties

Select a county

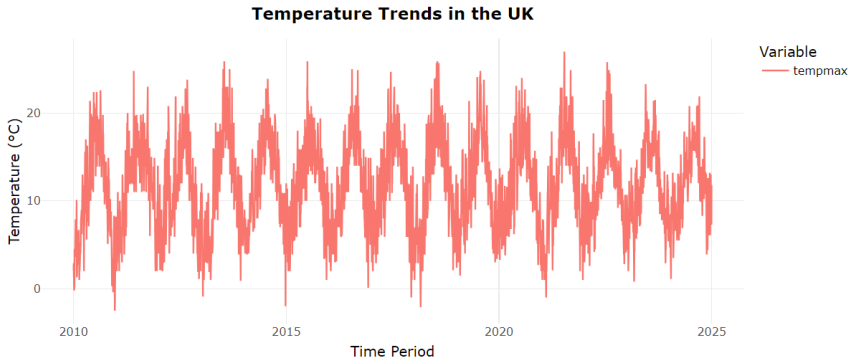
Aberdeen

Select Temperature Variables:

- [x] Maximum Temperature
- [ ] Minimum Temperature
- [ ] Average Temperature

Frequency Options:

- (•) Daily
- ( ) Monthly
- ( ) Yearly



Trends of Other Climate Variables in the UK Counties

Select a county

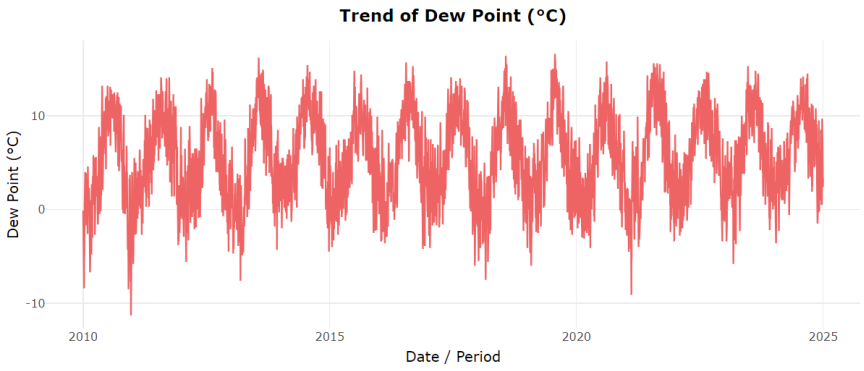
Aberdeen

Select Climate Variable:

Dew Point

Frequency Options:

- (•) Daily
- ( ) Monthly
- ( ) Yearly



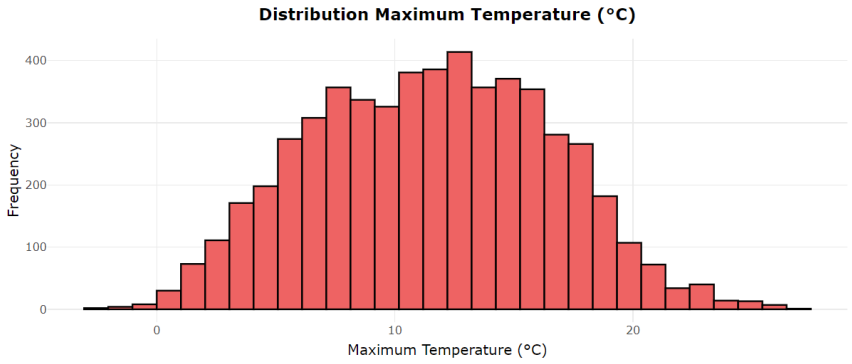
Distribution of Climate Variables in the UK Counties

Select a county

Aberdeen

Select Climate Variable:

Max Temp



Precipitation and Snow Over Time in the UK

Select a county

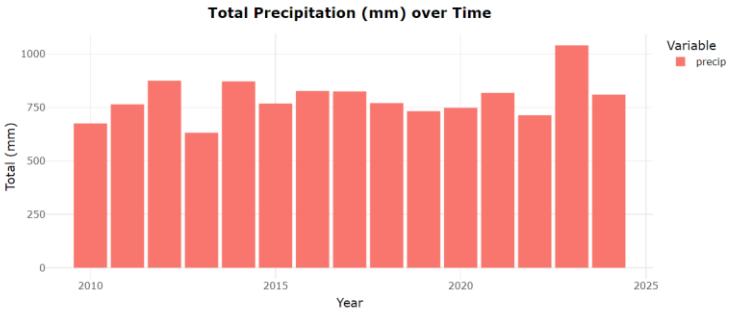
Aberdeen

Select Variables:

- Precipitation
- Snow

Frequency Options:

- Monthly
- Yearly



Risk Scatterplots

Select a county

Aberdeen

Select Scatter Plot:

Fog/Freezing Risk: Temperature vs Dew Point

Colour points by:

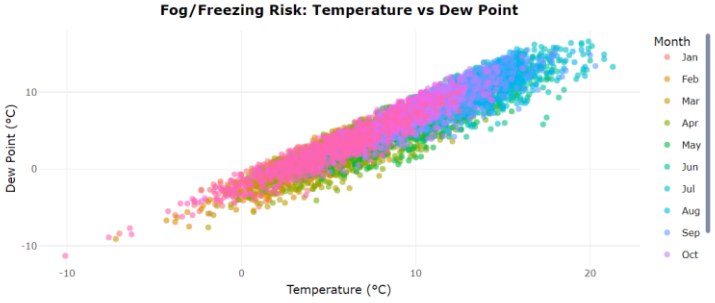
- Month
- Year

Filter by Month:

All

Filter by Year:

All



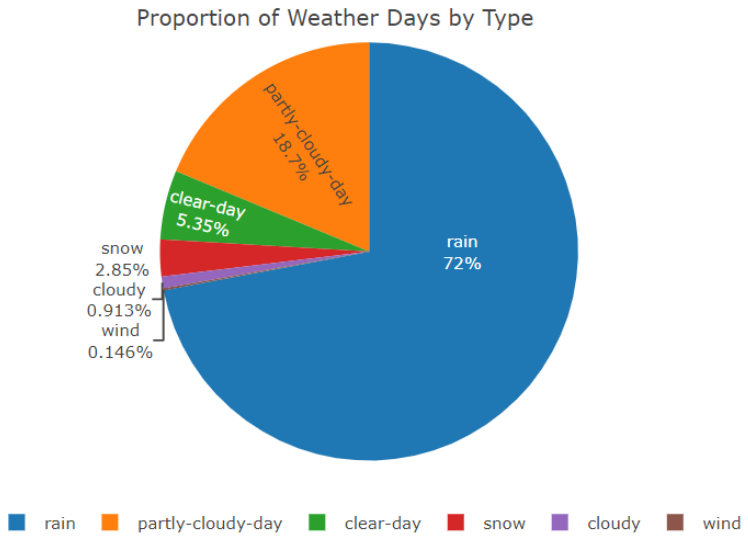
Proportion of Weather Days by Type

Select a county

Aberdeen

Year:

All



Top 100 Summary Table

Select a county

Aberdeen

Select Variable:

All

Select Year:

All

Show 10 entries

	name	datetime	tempmax	tempmin	temp	dew	humidity	precip	precipcover	snow	sno
1	Aberdeen	2021-07-17	26	15.9	21.3	13.3	62.1	0	0	0	
2	Aberdeen	2013-07-18	25.9	16	20.8	14.9	69.6	0.033	4.17	0	
3	Aberdeen	2018-07-15	25.7	16.9	20.8	14.3	67.2	0	0	0	
4	Aberdeen	2013-09-02	25	15	20.3	13.4	65.1	0	0	0	
5	Aberdeen	2019-07-	24.3	15.5	20.1	16	77.6	0	0	0	

MSc Project Presentation v2 - PowerPoint

Insurance Claims Dataset

Map | Graphs | Summary

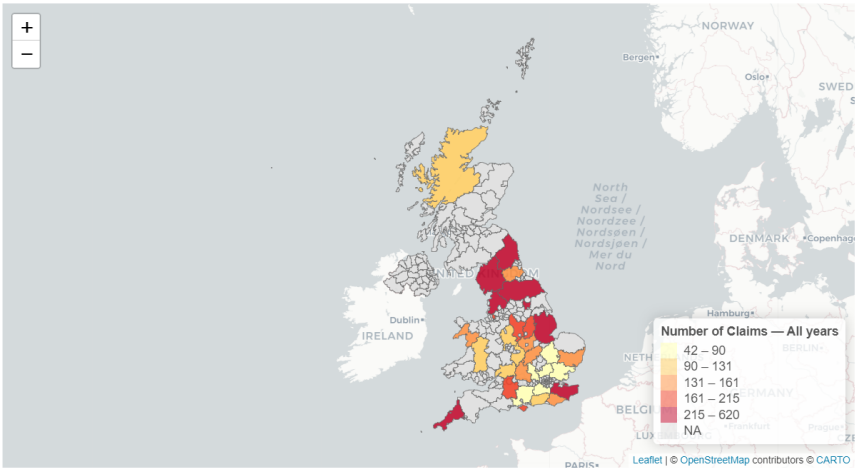
Insurance Claims — Map

Select Variable:

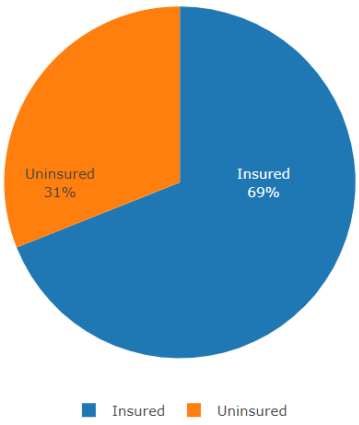
Number of Claims

Year:

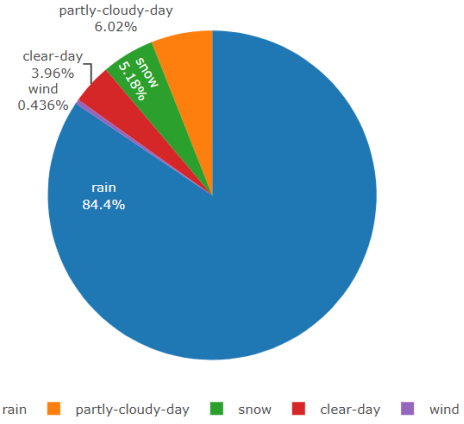
All



Proportion of Insured vs. Uninsured Claims



Claims by Description



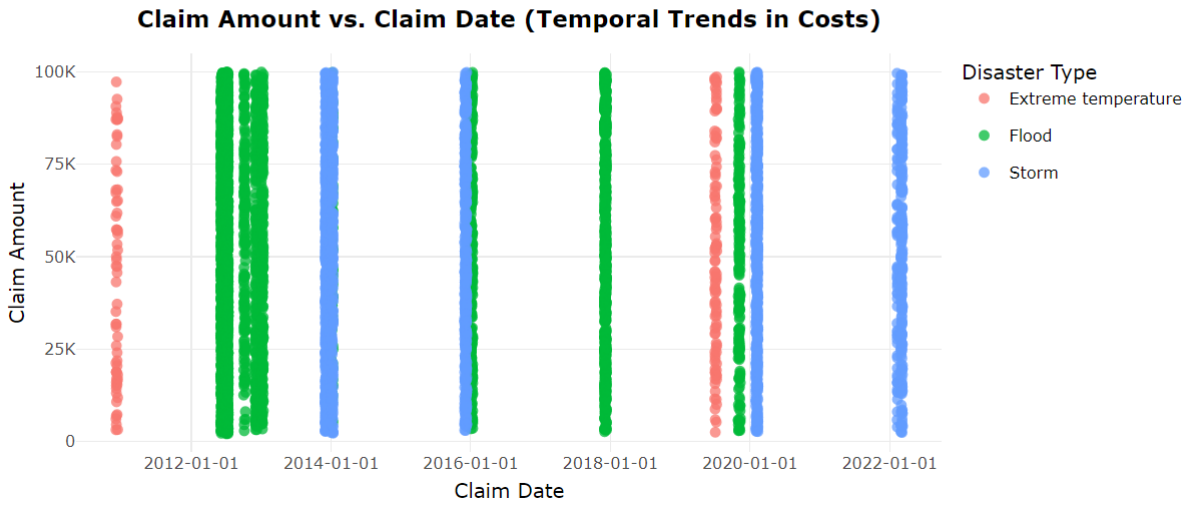
Claims: Scatterplots

Select Scatter Plot:

Claim Amount vs. Claim Date (Temporal Trends in Costs)

Filter by Disaster Type:

All



Grouped Summary Table (Natural Disasters)

Group by:

County

Show 10 entries

group	Total Claim Amount
North Yorkshire	30846700.6
Cumbria	27686001.21
Lancashire	25827011.6
Northumberland	21578861.74
Cornwall	13710703.77
Kent	12943043.89
Wiltshire	10850910.1
Isle of Wight	10810595.79

Natural Disaster Dataset

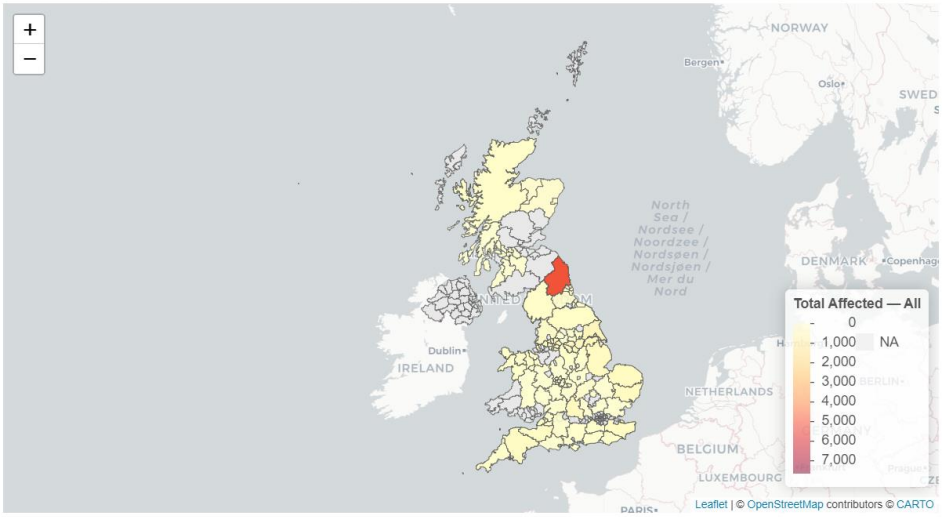
Natural Disasters — Choropleth Map

Select Variable:

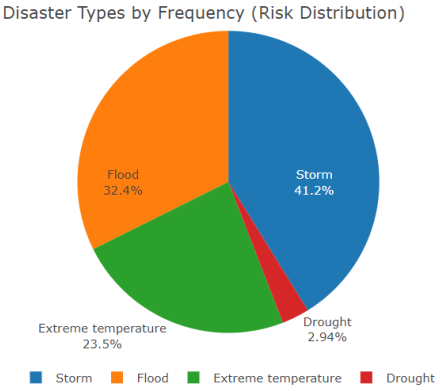
Total Affected

Year:

All



Disaster Types by Frequency



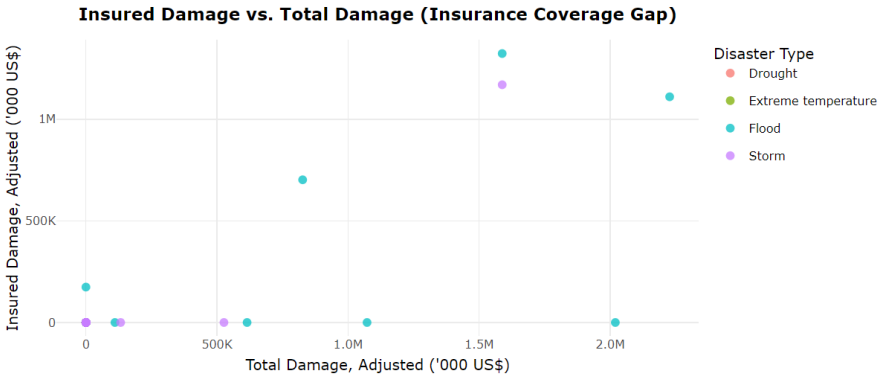
Risk Scatterplots for Natural Disasters)

Select Scatter Plot:

Insured Damage vs. Total Damage (Insurance Coverage Gap)

Select by Disaster Type:

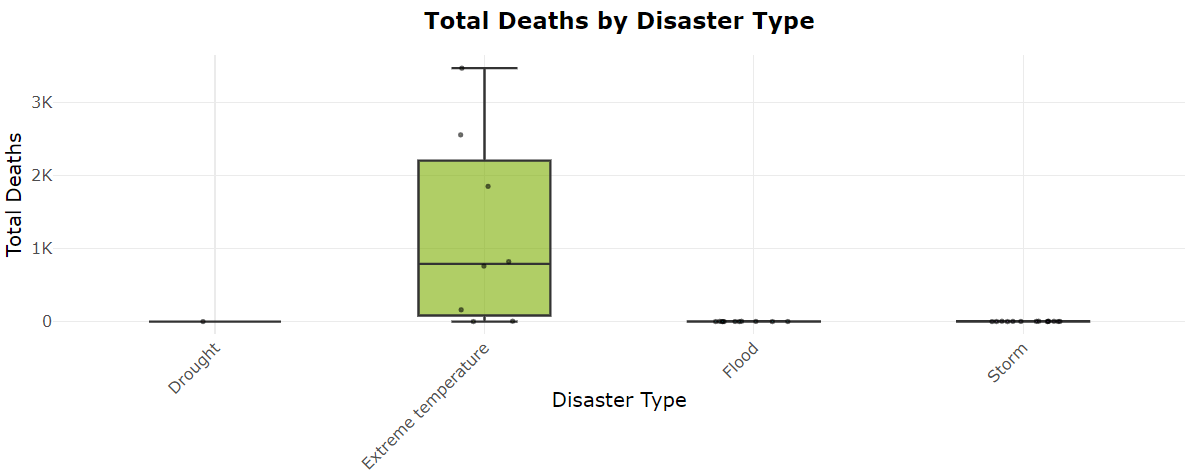
All



Risk Boxplots for Natural Disasters)

Select Box Plot:

Total Deaths



Grouped Summary Table (Natural Disasters)

Group by:

Disaster Type

Select Variable:

Total Affected

Show 10 entries

Disaster Type	Total Affected
Flood	53575
Storm	38554
Drought	0
Extreme temperature	0

Showing 1 to 4 of 4 entries

Previous 1 Next

Spatial Statistics

The spatial statistics utilises the Moran's I to compute the spatial diagnostics to get the counts, amounts and residuals.



Metric

Residual count (mean)

Year

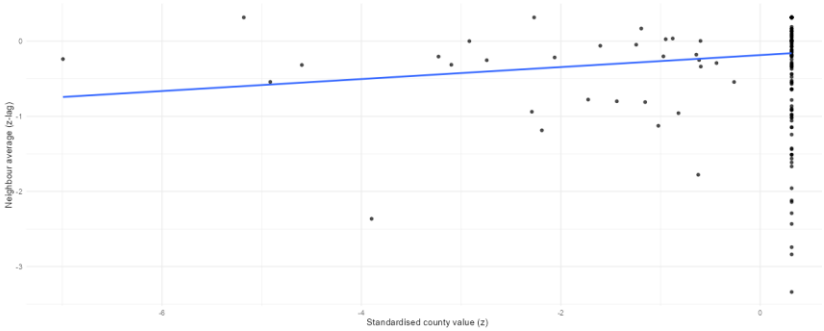
2024

Outputs: Global Moran's I (permutation p-value) and Moran scatterplot.

Global Moran's I

```
Moran's I = 0.07973   |   p-value = 0.048
```

Moran scatterplot



How to understand the results

Global Moran's I: close to +1 suggests clustering, near 0 suggests random, close to −1 suggests a checkerboard pattern. Use the permutation p value. p < 0.05 is commonly treated as evidence of non random structure

Moran scatterplot: x is the standardised county value, y is the average of neighbours. Quadrants: High–High, Low–Low, Low–High, High–Low. Points far from the origin are more influential.

Metric

Residual amount (mean)

Year

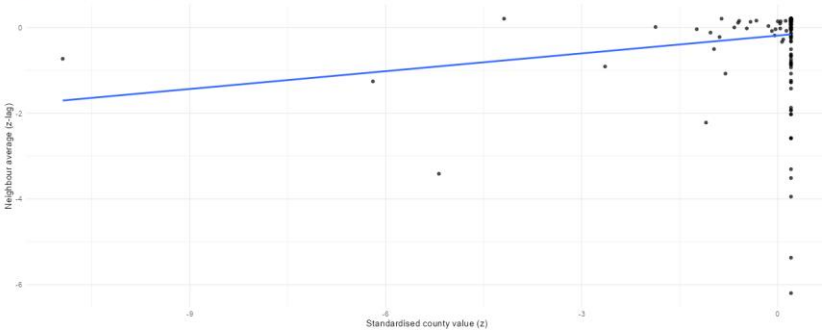
2024

Outputs: Global Moran's I (permutation p-value) and Moran scatterplot.

Global Moran's I

```
Moran's I = 0.1384   |   p-value = 0.018
```

Moran scatterplot



How to understand the results

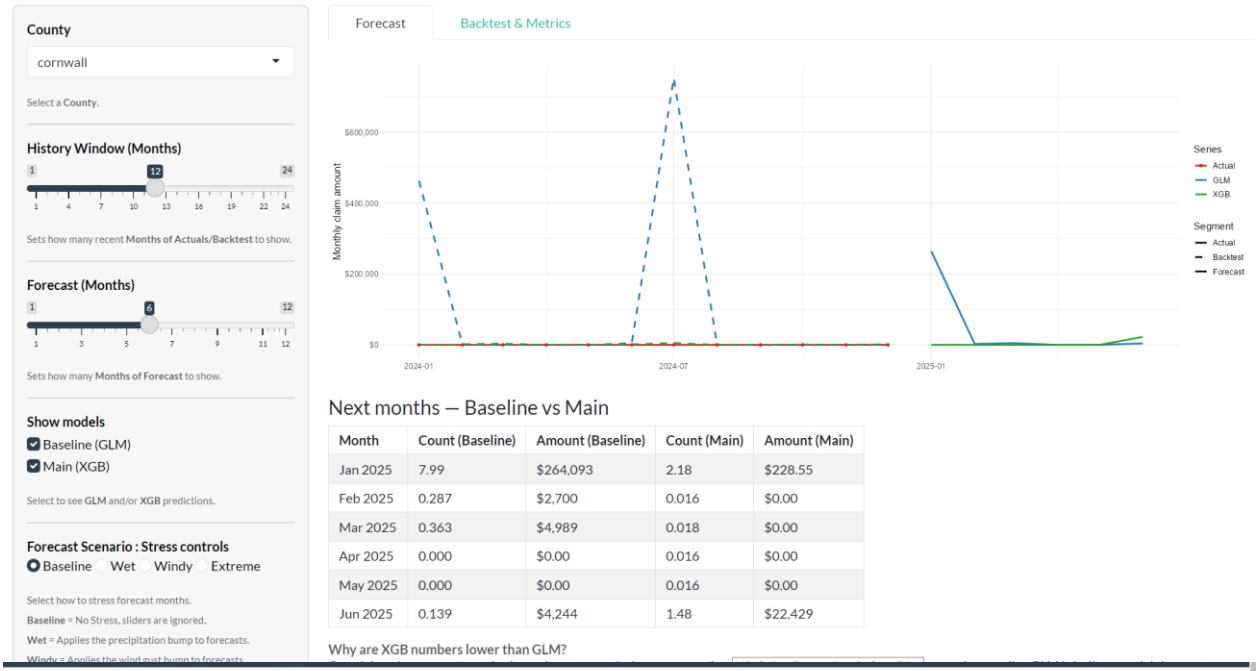
Global Moran's I: close to +1 suggests clustering, near 0 suggests random, close to −1 suggests a checkerboard pattern. Use the permutation p value. p < 0.05 is commonly treated as evidence of non random structure

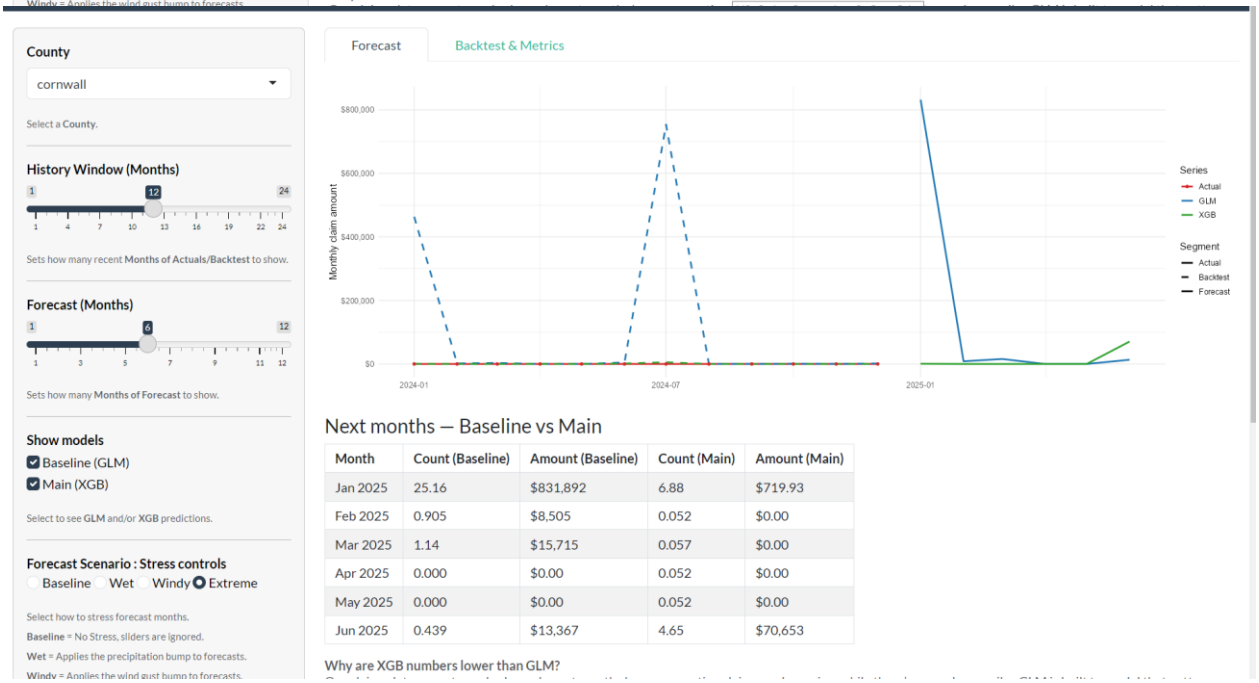
Moran scatterplot: x is the standardised county value, y is the average of neighbours. Quadrants: High–High, Low–Low, Low–High, High–Low. Points far from the origin are more influential.

Residual metrics: when a residual metric is selected, positive values mean the model under predicted, negative values mean it over predicted. Spatial structure in

Predictive Insights

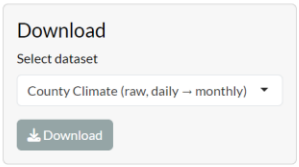
The predictive insights utilise the monthly panel and forecast the counts (Poisson) and amounts (Tweedie) using the GLM and XGBoost





About & Download

This tab show two thing that are reletaed to this app, first the explanation regarding this app and the second is downloading the raw and processed output in this app.



About This App

This application allows you to investigate climate data, records of natural disasters, and simulated insurance claims at the county level in the UK, as well as produce short-term forecasts for claim counts and amounts. The controls are user-friendly, the charts are clearly labelled, and the methods are thoroughly explained for better understanding.

What you can do

1. **Dashboard Overview**
   A summary overview displaying recent climate conditions, disaster events, and insurance claims at the national level, illustrated with key figures and trend visuals.
2. **Data Explorer**
   Deep dives into history.
   - **Maps**: Choropleth maps at the county level displaying climate indicators, insurance claims, and disaster impacts for specific years and variables.
   - **Graphs**: Flexible visualisations such as time series, histograms, bar charts, and scatter plots, featuring the ability to change counties and modify the time scale (daily, monthly, yearly).
   - **Summary**: Interactive tables of the underlying data.
3. **Predictive Insights**
   County-level forecasts.
   - Toggle **Baseline (GLM)** and **Main (XGB)** models.\
   - Choose history window and forecast horizon.\
   - Use scenario stress controls to adjust future precipitation, wind, and optional disaster lift.\
   - Review backtest metrics: MAE, WAPE, RMSE, bias ratio, zero-hit rate.
4. **Spatial Statistic**
   Assess spatial autocorrelation for a chosen metric and year.
   - **Global Moran's I** with permutation p-value.\
   - **Moran scatterplot** showing standardised county value vs neighbour average.\
   - Metrics include options such as residual count (mean) when selected.
5. **About & Download**
   Read this guide and download datasets.

Dataset	Path	Exists	Size
County Climate (raw, daily → monthly)	data/county_climate_refined.csv	Yes	130.6 MB
UK Climate (raw)	data/uk_climate_refined.csv	Yes	600 KB
Insurance Claims (raw)	data/Simulated_Insurance_Claims.csv	Yes	432.4 KB
Natural Disasters (raw)	data/natural_disaster_refined.csv	Yes	10.2 KB
Modeling Panel (monthly)	data/processed/ml_monthly_panel.csv	Yes	6.2 MB
Predictions (latest run)	data/processed/predictions_monthly.csv	Yes	1 MB
Validation Metrics (latest run)	data/processed/metrics_monthly.csv	Yes	391 B